



Probabilistic Modeling of User Interaction Patterns in Controlled Web Application Sessions

Transforming Web Sessions into Interpretable
Behavioral Models

Shaik Ibtihajulla Sha
Gaddala Abilash

This thesis is submitted to the Faculty of Computing at Blekinge Institute of Technology in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science. The thesis is equivalent to 10 Weeks of full-time studies.

The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

Contact Information:

Author(s):

Shaik Ibtihaajulla Sha

E-mail: ibsh25@student.bth.se

Gaddala Abilash

E-mail: abga25@student.bth.se

University advisor:

Oleksii Baranovskyi

Department of Computing

Faculty of Computing
Blekinge Institute of Technology
SE-371 79 Karlskrona, Sweden

Internet : www.bth.se
Phone : +46 455 38 50 00
Fax : +46 455 38 50 57

Abstract

Background. Understanding user interaction behavior in web applications requires modeling sequences of actions rather than isolated events. User sessions often contain structural patterns such as repeated navigation paths and transition dependencies between actions. However, access to large-scale interaction datasets is frequently restricted due to privacy and organizational limitations, making it difficult to study realistic behavioral patterns. This creates a need for lightweight and interpretable approaches that can model user behavior effectively under limited data conditions.

Objectives. This thesis investigates how effectively first-order and second-order Markov models can represent user navigation behavior in controlled web application sessions, what accuracy–robustness trade-offs arise under small-scale data conditions, and whether Markov-generated synthetic sessions can reproduce structural patterns observed in real sessions.

Methods. A controlled web application was developed and instrumented to collect structured interaction data from 49 user sessions consisting of sequences of 10–20 distinct action types. The collected logs were transformed into ordered session-based action sequences suitable for probabilistic modeling. Two probabilistic sequence models were implemented: a first-order Markov model and a second-order Markov model. Transition probabilities were estimated using Laplace smoothing to reduce sparsity and avoid zero-probability transitions.

Model evaluation was performed using 5-fold cross-validation with metrics including log-likelihood, perplexity, and Kullback–Leibler divergence. In addition, 500 synthetic sessions were generated through probabilistic sampling from the learned transition distributions. The generated sessions were compared with real interaction data using transition analysis, session-length distributions, and behavioral pattern extraction.

Results. The second-order Markov model achieved higher predictive performance, with improved log-likelihood (−90.14 vs −99.28) and lower perplexity (7.10 vs 7.28) compared to the first-order model. Both values are well below the uniform baseline perplexity of 19, confirming that both models learn non-trivial sequential structure. However, the second-order model also showed significantly higher divergence from real transition distributions (KL divergence: 61.12 vs 21.52), indicating greater sensitivity to sparse observations and increased risk of overfitting.

Synthetic session analysis showed that the second-order model captured sequential dependencies more effectively, reducing transition divergence by approximately 3.4% compared to the first-order model. It should be noted that both absolute JSD values (0.51 and 0.53) are near the theoretical maximum of $\ln 2 \approx 0.693$ nats, meaning this reduction occurs within a regime of poor overall fidelity. Analysis of real interaction sessions revealed strong navigation-oriented behavior, relatively short sessions (mean length 9.4), and recurring transition loops such as:

$$home_view \rightarrow view_cart \rightarrow home_view$$

These results demonstrate that even small-scale controlled datasets contain identifiable structural interaction patterns that can be modeled probabilistically.

Conclusions. The findings demonstrate a clear trade-off between model complexity and robustness in small-scale datasets. While second-order Markov models capture richer sequential dependencies and improve predictive accuracy, they are more vulnerable to sparsity, instability, and overfitting. First-order models, although simpler, provide more stable and reliable estimates under limited data conditions.

Overall, the study shows that probabilistic Markov modeling can successfully identify meaningful user interaction patterns in controlled small-scale environments. Synthetic sessions show limited local realism but are not sufficiently faithful to replace real session data for behavioral analysis. The results highlight the importance of balancing model complexity with available data when modeling user behavior and generating synthetic interaction sessions.

Keywords: Markov models, user interaction modeling, probabilistic sequence modeling, web sessions, behavioral analysis, sequence analysis

Acknowledgments

We would like to express our deepest gratitude to our supervisor, Oleksii Baranovskyi, for his continuous guidance, encouragement, and invaluable support throughout this thesis project. His constructive feedback, insightful discussions, and academic expertise played a significant role in shaping the direction and quality of this research. We are sincerely thankful for the time, patience, and dedication he invested in guiding us throughout the entire process.

We would also like to extend our appreciation to Blekinge Institute of Technology for providing an inspiring academic environment, the necessary resources, and the opportunity to conduct this research as part of our studies in Computer Science.

In addition, we are grateful to all participants who volunteered their time and contributed to the data collection process. Their participation and cooperation were essential for the successful completion of this study.

Finally, we would like to thank our families, friends, and classmates for their encouragement, motivation, and support during the course of this work. Their confidence in us helped us remain focused and motivated throughout the challenges of this thesis journey.

Contents

Abstract	i
Acknowledgments	iii
1 Introduction	1
1.1 Background	1
1.2 Research Problem	3
1.3 Research Motivation and Contributions	4
1.4 Thesis Outline	5
2 Related Work	7
2.1 Web Workload Modeling	7
2.2 Session-Based User Behavior Modeling	8
2.3 Probabilistic Sequence Modeling	9
2.4 Deep Learning Approaches	11
2.5 Synthetic Data and Privacy-Preserving Behavioral Analysis	11
2.6 Summary and Research Gap	12
3 Method	13
3.1 Research Questions	13
3.2 Research Question Operationalization	13
3.3 Research Design	14
3.4 Data Collection	16
3.4.1 Web Application Design	16
3.4.2 Participants and Data Volume	16
3.5 Data Representation	17
3.5.1 Session Construction	17
3.5.2 Action Space and Vocabulary	17
3.5.3 Sequence Characteristics	18
3.6 Probabilistic Modeling	18
3.6.1 Modeling Framework	18
3.6.2 First-Order Markov Model	18
3.6.3 Second-Order Markov Model	19
3.6.4 Model Construction and Parameter Estimation	19
3.6.5 Sparsity and Smoothing	20
3.7 Model Evaluation	20
3.7.1 Cross-Validation Strategy	20
3.7.2 Evaluation Metrics	21

3.7.3	Rationale for Multiple Metrics	22
3.8	Synthetic Session Generation	22
3.9	Pattern Analysis	23
3.10	Linking Method to Research Questions	23
3.11	Validity and Reliability	24
3.11.1	Internal Validity	24
3.11.2	External Validity	24
3.11.3	Construct Validity	24
3.11.4	Limitations	24
3.12	Summary	25
4	Results and Analysis	27
4.1	Model Evaluation Results	27
4.1.1	Overview of Evaluation Metrics	27
4.1.2	Log-Likelihood	28
4.1.3	Perplexity	28
4.1.4	Unseen Transition Ratio	28
4.1.5	KL Divergence	29
4.2	Model Comparison	29
4.2.1	Stability Analysis	29
4.2.2	Accuracy Comparison	29
4.2.3	Sparsity Trade-off	30
4.2.4	Final Model Recommendation	30
4.3	Synthetic Session Generation	30
4.4	Pattern Analysis	31
4.4.1	Action Distribution	31
4.4.2	Transition Distribution Analysis	32
4.4.3	Session Length Distribution	32
4.4.4	N-gram Overlap Analysis	33
4.4.5	Transition Hotspot Analysis	34
4.4.6	Behavioral Insights from Dominant Navigation Paths	35
4.5	Summary of Findings	36
5	Discussion	37
5.1	Revisiting the Research Questions	37
5.1.1	RQ1: Effectiveness of Markov Models for User Navigation Representation	37
5.1.2	RQ2: Accuracy–Robustness Trade-off	38
5.1.3	RQ3: Realism of Synthetic Session Generation	38
5.2	The Accuracy–Robustness Trade-off in Context	39
5.3	Why Synthetic Sessions Fall Short	39
5.4	Behavioral Insights and Their Implications for System Design	40
5.4.1	Influence of Application Design	40
5.5	Ethical Considerations in Synthetic Data Generation	41
5.6	Validity, Limitations, and Confidence in the Findings	41
5.7	Positioning the Findings in the Broader Literature	42
5.8	Summary	42

6	Conclusions and Future Work	43
6.1	Summary of the Study	43
6.2	Conclusions	43
6.2.1	RQ1: Effectiveness of Markov Models for Navigation Representation	43
6.2.2	RQ2: Accuracy–Robustness Trade-off	44
6.2.3	RQ3: Realism of Synthetic Session Generation	44
6.2.4	On the Value of Descriptive Markov Analysis	44
6.3	Contributions of the Thesis	44
6.4	Limitations and Unanswered Questions	45
6.5	Future Work	45
6.6	Final Remarks	46
	References	49
A	Supplemental Information	51
A.1	Per-Fold Evaluation Results	51
A.2	Complete Action Vocabulary	52
A.3	Transition Hotspot Analysis	52
A.4	Data and Code Repository	53

1.1 Background

Modern web applications have become an integral part of everyday life, supporting a wide range of services such as e-commerce, online education, healthcare systems, and public administration platforms. As these systems grow in complexity and scale, understanding how users interact with them has become increasingly important. Rather than consisting of isolated or independent actions, user behavior typically unfolds as a sequence of interactions over time. These sequences form structured sessions that reflect how users navigate systems, make decisions, and complete tasks.

Modeling such sequential behavior is essential for multiple reasons. First, it enables developers and system designers to better understand how users engage with applications, allowing for improvements in usability and system efficiency. Second, it plays a crucial role in performance evaluation and workload modeling, where realistic simulations of user behavior are required to test system scalability and robustness. Third, it provides insights into behavioral patterns that can be used for personalization, anomaly detection, and predictive analytics [17, 23].

A user session can be viewed as an ordered sequence of *discrete actions* — such as page visits, clicks, or form submissions — that occur within a bounded period of interaction [24]. These sequences often exhibit structural patterns, including repeated transitions, loops, and dependencies between actions, as illustrated in Figure 1.1. Capturing these patterns requires models that can represent not only individual actions but also the probabilistic relationships between them.

Web applications play a central role in modern digital ecosystems, supporting services across diverse domains. As user expectations continue to rise, there is an increasing demand for systems that are not only functional but also efficient, scalable, and user-centered. Understanding user interaction behavior is therefore a key requirement for designing high-quality web applications.

Traditional workload modeling approaches often rely on predefined scripts or static request patterns. While these methods are useful for controlled testing scenarios, they fail to capture the dynamic and probabilistic nature of real user behavior [2, 16]. In reality, users interact with systems in non-deterministic ways, influenced by individual preferences, goals, and contextual factors. Research on information foraging further demonstrates that user navigation behavior follows structured, goal-directed patterns shaped by the perceived value of information [20].

Recent research has shifted toward session-based representations, where user interactions are modeled as sequences of actions over time [5, 23]. These sequences

Figure 1: Typical User Interaction Session in an E-commerce Flow



Figure 1.1: Example of a user interaction session represented as a sequence of actions over time.

reveal structural properties such as navigation paths, action dependencies, and behavioral loops. Modeling these properties allows for more realistic workload simulation and deeper behavioral insights.

One of the most widely used approaches for modeling sequential data is the Markov model. Markov models represent systems as states (actions) and transitions between them, with associated probabilities, as shown in Figure 1.2. In a first-order Markov model, the probability of the next action depends only on the current action. In a second-order model, the probability depends on the two preceding actions, allowing for more complex dependencies to be captured [13, 19].

Markov Chain Model of User Navigation

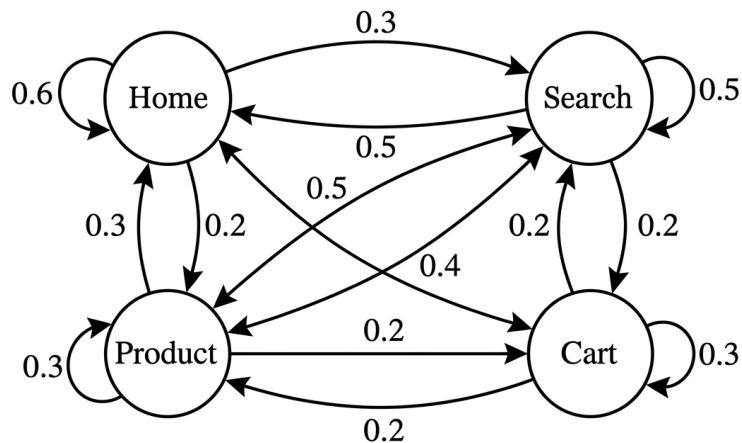


Figure 1.2: Illustration of a Markov model representing transition probabilities between user actions.

Advanced sequence modeling techniques, including deep learning approaches such as recurrent neural networks and transformer models, have demonstrated strong

performance in capturing long-range dependencies [9, 11]. However, these methods typically require large datasets and significant computational resources, making them less suitable for small-scale or constrained data environments.

In contrast, probabilistic models such as Markov chains provide a simpler and more interpretable framework for modeling sequential dependencies [19]. They are particularly well-suited for scenarios where data is limited, as they require fewer parameters and are easier to analyze. The theoretical properties of finite Markov chains, including convergence behavior and transition stability, are well established in the literature [14].

In this thesis, user interaction behavior is represented as sequences of discrete actions, and probabilistic dependencies between actions are modeled using first-order and second-order Markov models. These models enable both the analysis of behavioral patterns and the generation of synthetic interaction sessions that mimic real user behavior. Model performance is evaluated using standard sequence modeling metrics such as perplexity and log-likelihood [12].

1.2 Research Problem

This thesis investigates how effectively first-order and second-order Markov models can represent user navigation behavior, what accuracy–robustness trade-offs arise under small-scale data conditions, and whether Markov-generated synthetic sessions can reproduce structural patterns observed in real sessions.

The study is conducted in a controlled experimental setting, where interaction data is collected from 49 user sessions within a custom-built web application. As illustrated in Figure 1.3, the study follows a structured pipeline consisting of data collection, preprocessing, model construction, evaluation, and synthetic session generation. The session preprocessing step follows established practices in web usage mining [22], ensuring that sessions are correctly identified and structured before modeling.

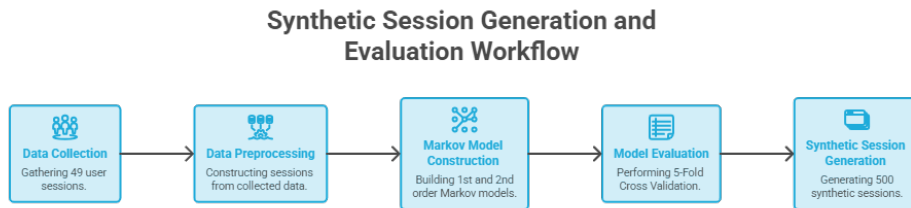


Figure 1.3: Overall workflow of the study, including data collection, modeling, evaluation, and synthetic session generation.

The scope of this thesis is bounded in three key aspects. First, the number of participants and sessions is relatively small, reflecting realistic constraints such as limited access to large-scale behavioral data. Second, the action space is restricted to a finite set of discrete actions, enabling reliable probabilistic estimation while maintaining interpretability [13]. Third, the analysis focuses on observable structural

patterns in action sequences, without explicitly modeling user intent or contextual influences.

The central problem addressed in this thesis is therefore:

To what extent can structural user interaction patterns be reliably modeled using first-order and second-order Markov models under small-scale, controlled data conditions?

To address this problem, the study compares the models in terms of predictive performance, transition stability, divergence from real data distributions, and their ability to generate realistic synthetic sessions [3]. This enables an empirical evaluation of the trade-off between model complexity and robustness under data constraints.

1.3 Research Motivation and Contributions

Understanding user interaction behavior is not only a technical challenge but also a practical necessity in modern system design. Many real-world systems operate under conditions where large-scale user data is unavailable due to privacy concerns, deployment constraints, or early-stage development limitations. Despite this, developers and researchers are still required to evaluate system performance, simulate realistic workloads, and derive meaningful insights from limited observations.

This creates a fundamental challenge: how to construct reliable behavioral models when data is sparse. In such scenarios, overly complex models may overfit the available data, capturing noise rather than meaningful structure. Conversely, overly simplistic models may fail to capture essential behavioral patterns, leading to inaccurate representations of user interactions [6].

Probabilistic models, particularly Markov models, offer a balanced solution to this challenge. They provide a mathematically grounded framework for modeling sequential dependencies while maintaining interpretability and computational efficiency [19]. By explicitly representing transition probabilities between actions, these models enable both descriptive and generative analysis of user behavior. Empirical studies on web navigation confirm that Markov-based models are effective for capturing sequential clickstream patterns [4, 6, 18].

Another important motivation for this study lies in the generation of synthetic user sessions. In many cases, real user data cannot be freely shared due to ethical and privacy considerations [7, 8]. Synthetic data generation provides an alternative approach, allowing researchers to create realistic datasets that preserve structural properties of user behavior without exposing sensitive information. However, the quality of such synthetic data depends heavily on the underlying behavioral model [3].

The main contributions of this thesis are as follows:

- An empirical comparison of first-order and second-order Markov models applied to controlled web session data, quantifying how model order affects predictive accuracy, distributional alignment, and sparsity under small-scale data conditions [6, 13, 19].

- A cross-validated evaluation framework using four complementary metrics — log-likelihood, perplexity, KL divergence, and unseen transition ratio — that together expose the accuracy–robustness trade-off between model orders [12].
- A multi-dimensional assessment of synthetic session fidelity, combining JSD-based transition analysis, session length statistics, n-gram overlap, and transition hotspot identification [3, 7, 8].
- Concrete behavioral insights derived from the real session corpus, including dominant navigation paths, transition hotspots, and recurring interaction loops [18, 24].
- A reproducible end-to-end pipeline for controlled user behavior data collection, Markov-based sequence modeling, and synthetic session generation that can be adapted to other bounded web application environments.

1.4 Thesis Outline

This thesis is organized into six chapters, each addressing a specific aspect of the study.

Chapter 2 presents an overview of existing research on user interaction modeling, workload generation, and probabilistic sequence models. Chapter 3 details the experimental design, data collection process, and modeling techniques used in this study.

Chapter 4 presents the evaluation results, including model performance metrics, synthetic session analysis, and pattern comparisons between real and generated data. Chapter 5 interprets these findings, highlighting key insights such as dominant navigation patterns, transition hotspots, and model limitations.

Finally, Chapter 6 summarizes the contributions of the thesis and discusses potential avenues for future research in user behavior modeling under constrained data conditions.

Modeling user interaction behavior in web applications has been widely studied across multiple research domains, including performance engineering, human-computer interaction, data mining, and machine learning. As modern web systems have evolved into highly dynamic and interactive platforms, understanding how users navigate and interact with these systems has become increasingly important for both system optimization and user experience design [5, 23].

User behavior modeling serves several key purposes. It supports realistic workload generation for performance testing, enables behavioral analysis for system improvement, and provides the foundation for predictive models used in recommendation systems and personalization. However, accurately capturing user behavior remains a challenging task due to its inherently sequential, stochastic, and context-dependent nature [20].

Existing approaches to modeling user behavior can be broadly categorized into three main groups: traditional workload modeling techniques, session-based behavioral modeling approaches, and advanced sequence modeling methods. Each of these approaches differs in terms of complexity, data requirements, interpretability, and suitability for different application contexts.

This chapter reviews the most relevant literature in these areas and positions this thesis within the broader research landscape. In particular, it highlights the limitations of existing approaches when applied to small-scale datasets and motivates the use of probabilistic sequence models as a balanced solution.

2.1 Web Workload Modeling

Early research in web workload modeling primarily focused on generating synthetic traffic to evaluate system performance under different load conditions. These approaches were designed to simulate user requests at the system level, often without explicitly modeling the underlying user behavior.

Barford and Crovella [2] proposed statistical techniques for generating representative web workloads based on empirical observations of real traffic. Their work emphasized capturing aggregate properties such as request inter-arrival times, file size distributions, and server response characteristics. This approach provided valuable insights into how realistic traffic patterns affect system performance.

Similarly, Menascé and Almeida [16] introduced analytical models for capacity planning in web systems. Their work focused on predicting system performance metrics such as throughput, response time, and resource utilization under varying

workloads. These models are particularly useful for infrastructure planning and system scaling.

Despite their contributions, traditional workload modeling approaches have notable limitations. One of the primary shortcomings is the assumption that user requests can be treated as independent or loosely correlated events. In reality, user interactions are highly structured and sequential, with each action influenced by previous actions within a session [18].

In practical applications, tools such as Apache JMeter [1] and Locust [15] are widely used for performance testing. These tools allow developers to define custom scenarios and simulate concurrent users. However, they rely heavily on manually scripted workflows, which introduces several challenges. First, creating realistic scripts requires domain expertise and significant effort. Second, predefined scripts cannot easily adapt to diverse or unpredictable user behaviors. Third, they fail to capture emergent patterns such as loops, branching navigation paths, and probabilistic transitions.

As a result, while traditional workload modeling is effective for system-level testing, it does not adequately represent the structural and sequential nature of real user interactions. This limitation motivates the need for more behaviorally grounded modeling approaches.

2.2 Session-Based User Behavior Modeling

To overcome the limitations of traditional workload models, researchers have increasingly adopted session-based representations of user behavior. In this approach, user interactions are modeled as ordered sequences of actions, where each sequence represents a complete session of user activity within a system [23].

The concept of web usage mining, introduced by Cooley et al. [5], established a formal framework for discovering behavioral patterns from server log data. This work laid the foundation for session-based analysis by defining how raw clickstream data can be preprocessed into structured interaction sequences suitable for pattern discovery. Building on this, Mobasher et al. [17] demonstrated how session-level usage patterns can be leveraged for automatic web personalization, highlighting the practical value of structured sequence modeling. These studies directly motivate the session-based representation adopted in this thesis. Rather than treating user actions as isolated events, interaction logs are transformed into ordered action sequences so that navigation behavior can be analyzed as a structured process over time.

Session-based modeling provides a more realistic and fine-grained view of user behavior. By preserving the temporal order of actions, it becomes possible to analyze how users navigate through a system, how decisions are made at each step, and how different interaction paths emerge. Studies of clickstream data confirm that user navigation follows structured, non-random paths with identifiable transition patterns [18, 24].

Session-based representations reveal several important structural properties of user behavior, as illustrated in Figure 2.1. These include frequently occurring navigation paths, repeated transitions between specific actions, and behavioral loops where users revisit the same states multiple times. Additionally, session data can

highlight drop-off points where users exit the system, providing insights into potential usability issues [20].

Figure 1: User Session Path Patterns

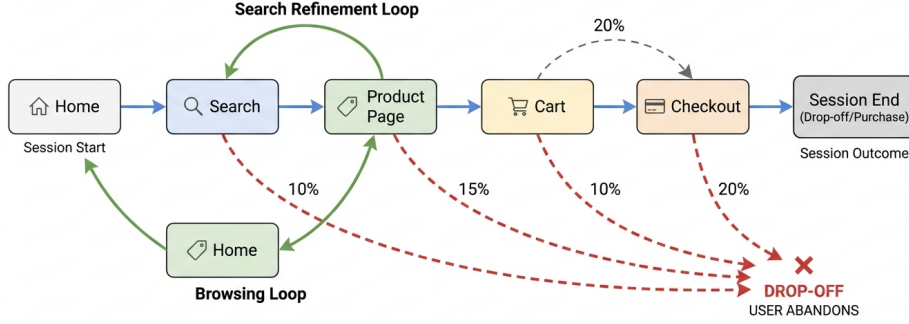


Figure 1: User Session Path Patterns

Figure 2.1: Illustration of common session-based behavioral patterns, including navigation paths, repeated transitions, and interaction loops.

A key preprocessing challenge in session-based modeling is the accurate reconstruction of user sessions from raw log data. Spiliopoulou et al. [22] proposed a systematic framework for evaluating session reconstruction heuristics, addressing issues such as session boundary detection, user identification, and noise filtering. This work is directly relevant to the preprocessing pipeline used in this thesis.

However, session-based modeling also introduces challenges. One key issue is data sparsity, particularly in small datasets where many possible action sequences may be observed only a few times or not at all. This makes it difficult to estimate reliable transition probabilities and limits the ability to generalize beyond observed data [6].

Another challenge is the trade-off between granularity and interpretability. While detailed session representations capture rich behavioral information, they can also become complex and difficult to analyze without appropriate modeling techniques. This highlights the need for models that can effectively summarize and generalize sequence data while preserving essential structural patterns [4].

2.3 Probabilistic Sequence Modeling

Probabilistic sequence models provide a principled framework for representing and analyzing sequential data. Among these, Markov models are particularly well-suited for modeling systems where the probability of future states depends on a limited history of past states.

Norris [19] provides the theoretical foundation for Markov chains, formalizing the concept of the Markov property. In a first-order Markov model, the next state depends only on the current state, which simplifies the modeling process and reduces the number of parameters. In contrast, higher-order Markov models incorporate additional historical context, allowing for more complex dependencies to be captured.

The mathematical properties of finite Markov chains, including absorbing states and stationary distributions, are further formalized by Kemeny and Snell [13], providing the algebraic tools needed to analyze transition matrices over finite action spaces.

Levin et al. [14] extend the theoretical treatment of Markov chains to cover convergence rates and mixing times, which are relevant when assessing the stability of estimated transition matrices under data constraints. In practical terms, this relates to how quickly a model’s estimated behavior converges to its true distribution as more session data is observed.

The choice of model order introduces an important trade-off. Lower-order models are more robust and require less data, but they may fail to capture important dependencies between actions. Higher-order models are more expressive but require significantly more data to estimate parameters reliably. This can lead to sparsity issues, particularly in small datasets [6]. Borges and Levene [4] investigated this trade-off specifically in the context of web navigation, demonstrating that variable-length Markov models can balance expressiveness and coverage better than fixed higher-order models when data is limited. Their empirical results provide a direct reference point for interpreting the KL divergence and n-gram overlap findings reported in Chapter 4. The findings of Borges and Levene are particularly relevant to this study because they highlight the trade-off between model expressiveness and data availability. This thesis investigates a similar question by comparing first-order and second-order Markov models under limited data conditions and examining how increased model complexity affects predictive performance and robustness. Rabiner [21] extended the Markov framework through Hidden Markov Models (HMMs), which introduce latent states to model underlying processes that are not directly observable. HMMs have been successfully applied in domains such as speech recognition and bioinformatics. However, their increased number of parameters and reduced interpretability make them less suitable for a small-scale dataset of 49 sessions, where reliable estimation of latent state emission distributions would require substantially more data than is available in this study.

In the context of user behavior modeling, Markov models offer several advantages. They provide a clear and interpretable representation of transition probabilities between actions, making it easier to analyze behavioral patterns [24]. They are also computationally efficient and can be trained with relatively small datasets. Clickstream studies confirm that even first-order models capture dominant navigation paths well, while second-order extensions yield meaningful improvements in predictive accuracy [6, 18].

Evaluating sequence models requires appropriate metrics. Jurafsky and Martin [12] provide a comprehensive treatment of perplexity and log-likelihood as standard measures of probabilistic model quality, adapted here from language modeling to user interaction sequences.

This thesis focuses on first-order and second-order Markov models as a means of balancing simplicity and expressiveness. By comparing these models, it becomes possible to evaluate how additional historical context influences predictive performance and the ability to generate realistic interaction sequences.

2.4 Deep Learning Approaches

Recent advances in machine learning have led to the widespread adoption of deep learning techniques for sequence modeling. Methods such as recurrent neural networks (RNNs), long short-term memory (LSTM) networks, and transformer-based architectures are capable of capturing complex, long-range dependencies in sequential data.

Goodfellow et al. [9] provide a comprehensive overview of deep learning methods, highlighting their ability to learn hierarchical representations from large datasets. In the context of user behavior modeling, these techniques have been applied to tasks such as next-click prediction, recommendation systems, and user intent modeling. Hidasi et al. [11] demonstrated that gated recurrent units (GRUs) can effectively model session-based interaction sequences for recommendation, achieving strong predictive performance on large-scale datasets. This line of work represents the state of the art in deep learning for session modeling; however, its data requirements — typically thousands to millions of sessions — make it unsuitable for the controlled 49-session setting examined in this thesis.

In large-scale industrial applications, deep learning models have demonstrated strong predictive performance. For example, He et al. [10] describe practical implementations of deep learning systems for user behavior prediction, leveraging massive datasets to achieve high accuracy.

One of the primary challenges for deep learning is the requirement for large amounts of training data. Deep learning models typically contain a large number of parameters, making them prone to overfitting when data is limited [9]. Additionally, they require significant computational resources for training and tuning, and unlike probabilistic models they do not provide explicit representations of transition probabilities, making it difficult to extract interpretable behavioral insights [11].

In controlled experimental settings with limited data, these limitations become particularly significant. Simpler and more interpretable models are therefore more appropriate for the scenario investigated in this thesis. These considerations motivate the decision to focus on Markov models in the present study. Given the limited dataset of 49 sessions, model interpretability and robustness are prioritized over the representational capacity offered by deep learning approaches.

2.5 Synthetic Data and Privacy-Preserving Behavioral Analysis

A growing body of research addresses the generation of synthetic behavioral data as an alternative to sharing real user interaction logs, which may be subject to privacy regulations and ethical constraints.

Gonzalez et al. [8] propose PrivGen, a Markov model-based approach to synthetic sequential data generation that preserves structural properties of the original data while protecting individual privacy. This work is directly parallel to the synthetic session generation component of this thesis, and confirms that probabilistic models are well-suited for generating realistic behavioral sequences under data constraints.

Their results are used as a qualitative reference when interpreting the distributional fidelity of the synthetic sessions reported in Chapter 4.

Bindschaedler and Shokri [3] introduce the concept of plausible deniability for evaluating the quality of synthetic data, providing a principled framework for assessing how closely synthetic records resemble real ones. This offers a useful lens for interpreting the distributional comparison between synthetic and real sessions presented in this thesis.

El Emam et al. [7] provide a broader treatment of synthetic data generation methods, covering the trade-offs between data utility and privacy protection. Their work reinforces the motivation for generating synthetic sessions as a means of enabling behavioral research without exposing sensitive user records.

2.6 Summary and Research Gap

The literature reviewed in this chapter highlights a clear progression in user behavior modeling approaches, from simple workload generation techniques to advanced sequence modeling methods. While traditional approaches are efficient and widely used in practice [2, 16], they fail to capture the sequential structure of user interactions. Session-based models improve upon this by treating user behavior as ordered sequences [5, 23], but require careful preprocessing and are sensitive to data sparsity [22]. On the other hand, modern deep learning methods offer high expressive power [9, 11] but require large-scale datasets and significant computational resources.

This creates a gap in the existing research: there is a lack of approaches that can effectively model sequential user behavior while remaining robust and interpretable under small-scale data conditions.

This thesis addresses this gap by focusing on probabilistic sequence modeling using first-order and second-order Markov models [4, 6, 19]. By operating in a controlled experimental setting with limited data, the study provides insights into how model complexity influences performance, stability, and realism.

Furthermore, by incorporating synthetic session generation [7, 8] and pattern analysis, this work extends beyond traditional evaluation metrics to assess how well the models capture the structural properties of user behavior. This contributes to a deeper understanding of sequence modeling in constrained environments and provides a foundation for future research in this area.

This chapter describes the methodological framework used to investigate how user interaction sequences in web applications can be modeled using probabilistic approaches under constrained data conditions. The methodology is structured as a systematic pipeline that progresses from raw data collection through preprocessing, probabilistic modeling, cross-validated evaluation, synthetic session generation, and behavioral pattern analysis. Each stage is designed to build upon the previous one, ensuring that the analytical process remains coherent, transparent, and reproducible from end to end.

Special care is given to documenting design decisions that were made in response to the known constraints of the dataset, particularly its limited size, so that the reader can assess the validity of the subsequent results in proper context. All steps were implemented in Python using standard scientific libraries to ensure portability and reproducibility.

3.1 Research Questions

The study is guided by three research questions, each of which targets a distinct layer of the investigation.

- **RQ1:** How effectively do first-order and second-order Markov models represent user navigation behavior in controlled web application sessions?
- **RQ2:** What trade-offs exist between predictive accuracy and robustness when applying first-order and second-order Markov models to small-scale user interaction datasets?
- **RQ3:** To what extent can synthetic sessions generated from Markov models reproduce the structural interaction patterns observed in real user sessions?

3.2 Research Question Operationalization

Translating abstract research questions into concrete, measurable procedures is a necessary step for ensuring that the study produces interpretable and verifiable results. Each research question is operationalized through specific technical steps that produce quantifiable outputs, creating a direct line of traceability from theory to implementation.

RQ1 is operationalized by constructing both a first-order and a second-order Markov model from the collected session sequences and evaluating how well each model captures the navigation structure of the real data. Specifically, representational effectiveness is assessed through the learned transition matrices, perplexity relative to the uniform baseline, and KL divergence from the empirical distribution. RQ2 is operationalized by comparing the two models across all four cross-validation metrics — log-likelihood, perplexity, KL divergence, and unseen transition ratio — with explicit attention to the sparsity of the second-order state space and its consequences for distributional alignment. RQ3 is operationalized through a post-generation pattern analysis that examines action distributions, transition JSD, session length statistics, n-gram overlaps, and transition hotspots between real and synthetic sessions. The connection between each research question and its corresponding analysis is summarized in Table 3.1.

Table 3.1: Mapping of research questions to methodological steps and evaluation measures.

RQ	Methodological step	Evaluation measure
RQ1	First-order and second-order Markov model training; transition matrix analysis	Perplexity vs. uniform baseline, KL divergence, transition coverage
RQ2	Cross-validated model comparison	Log-likelihood, perplexity, KL divergence, unseen transition ratio, sparsity analysis
RQ3	Synthetic session generation and pattern analysis	JSD, n-gram overlap, session length statistics, transition hotspots

3.3 Research Design

The study adopts a quantitative experimental design in which the primary independent variable is model complexity, instantiated as the choice between a first-order and a second-order Markov model. The dependent variables are the evaluation metrics described in Section 3.7, namely log-likelihood, perplexity, KL divergence, and unseen transition ratio. This design makes it possible to isolate the effect of adding one additional step of sequential memory on model fit, generalization, and behavioral realism.

A controlled web application environment was used for data collection in order to eliminate the confounding effects that arise from unstructured, naturalistic browsing. By constraining the interaction space to a predefined set of actions and a defined application flow, the study ensures that all sessions are drawn from the same underlying behavioral process, which is a prerequisite for valid probabilistic estimation. The overall research workflow consists of six sequential phases:

1. Data collection via the instrumented web application
2. Preprocessing and session segmentation
3. Probabilistic model training (first-order and second-order Markov)
4. Cross-validated evaluation using quantitative metrics
5. Synthetic session generation by probabilistic sampling
6. Pattern analysis comparing real and synthetic sessions

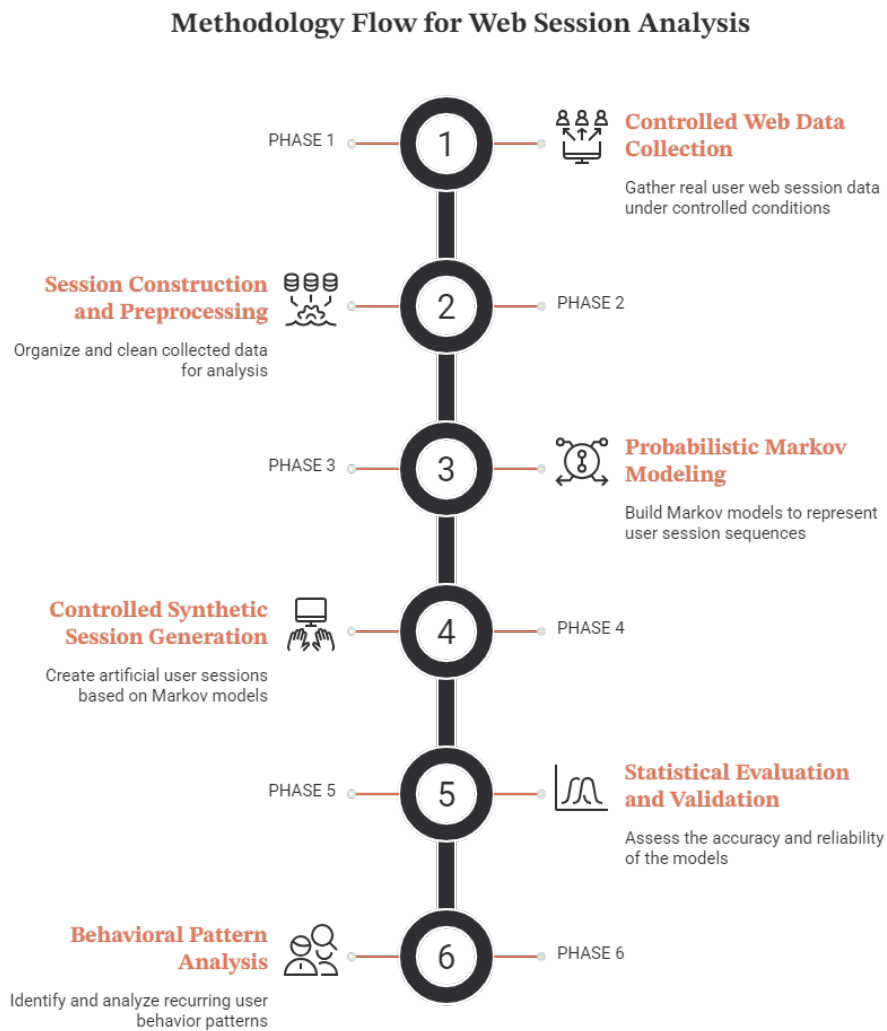


Figure 3.1: Overall methodology pipeline illustrating the six phases from data collection through pattern analysis.

3.4 Data Collection

3.4.1 Web Application Design

A custom web application was developed specifically for this study to enable the controlled collection of structured interaction data. The application simulates a lightweight digital bookshop that offers users the ability to browse stories by genre, view individual story pages, read previews, add items to a cart, and complete a check-out process. This design was deliberately chosen because it replicates a realistic and familiar interaction flow, which reduces the likelihood of artificial or exploratory behavior caused by confusion with the interface. At the same time, the bounded action space ensures that all user-generated events belong to a well-defined vocabulary that can be modeled probabilistically.

Every user event was captured by a client-side logging module that recorded three fields for each interaction: a session identifier generated as a random browser cookie value, an action type label drawn from a predefined taxonomy of 19 distinct actions (including **START** and **END** boundary tokens), and a Unix timestamp with millisecond resolution. Logs were transmitted to a lightweight server-side endpoint and stored in a structured format for subsequent processing. The logging module was validated through pilot testing prior to formal data collection to ensure completeness and accuracy of the recorded events.

3.4.2 Participants and Data Volume

A total of 49 sessions were collected from 34 voluntary participants. Some participants contributed more than one session in order to reach the target session count; the number of sessions per participant ranged from one to three. Participants were invited to interact with the web application under a combination of free exploration and task-directed scenarios. All participants received the same general instructions and were encouraged to use the application as they would a typical online bookshop. To ensure that all major features were exercised, participants were asked to browse genres, open story pages, read previews, add items to the cart, view the cart, and complete the checkout process where appropriate. However, no fixed navigation path or strict workflow was imposed. Participants were free to decide how they interacted with the system, allowing natural variations in navigation behavior while ensuring that all core application features were represented in the collected sessions.

Where a participant contributed multiple sessions, those sessions were kept together in the same cross-validation fold to minimize within-participant leakage across the training-validation boundary. The 49 sessions represent the complete dataset used for all subsequent modeling and analysis steps, with no sessions excluded after collection..

3.5 Data Representation

3.5.1 Session Construction

Each raw interaction log was segmented into individual sessions using the session identifier as the grouping key. Within each session, events were sorted by timestamp to produce an unambiguous temporal order. The resulting session is represented as an ordered sequence of action labels:

$$S = \langle a_1, a_2, \dots, a_n \rangle$$

where each a_i is drawn from the action vocabulary $\mathcal{A}$. Two special tokens are appended to every sequence: a **START** token prepended before the first observed action and an **END** token appended after the last observed action. These boundary tokens serve a dual purpose: they allow the model to learn the probability distribution over initial actions (i.e., $P(a_1 \mid \text{START})$) and they provide a natural termination condition during synthetic session generation. Without explicit boundary tokens, the model would have no principled way to reproduce the observed frequency of session starts and ends.

An example session sequence with boundary tokens is illustrated in Figure 3.2.

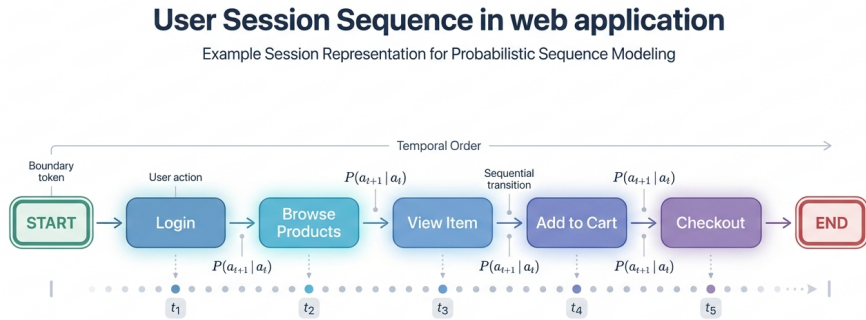


Figure 3.2: Illustration of an example session sequence with **START** and **END** boundary tokens. Actions are shown in temporal order from left to right.

3.5.2 Action Space and Vocabulary

The complete action vocabulary observed across all 49 sessions consists of 19 distinct action types, including the two boundary tokens. The substantive actions correspond to distinct navigational and transactional events within the web application, such as `home_view`, `select_genre_fantasy`, `view_story_fantasy`, `read_preview`, `add_to_cart`, `view_cart`, `checkout`, and `session_end`, among others. A vocabulary of this size is sufficient to capture the behavioral richness of the application

while remaining tractable for probability estimation from a dataset of 49 sessions. Actions are mapped to integer indices for efficient matrix and tensor operations; this encoding is strictly a computational convenience and carries no ordinal meaning.

3.5.3 Sequence Characteristics

An analysis of the raw sequences prior to modeling reveals several structural properties that directly inform the modeling choices made in this study. Sessions vary substantially in length, with a mean of 9.37 actions and a maximum of 27 actions, indicating that users follow qualitatively different interaction patterns rather than a single stereotyped path. The data exhibits repeated transitions across sessions, suggesting that certain behavioral pathways are strongly preferred. Navigation loops, in which a user returns to a previously visited state, are also present in the real data and represent an important structural feature that the models should ideally reproduce.

3.6 Probabilistic Modeling

3.6.1 Modeling Framework

The modeling framework adopted in this study is based on finite-order Markov chains, which provide a mathematically tractable and interpretable approach to sequential dependency estimation. A Markov chain of order k assumes that the probability of the next action depends only on the k most recent actions, not on the full history of the session. This assumption makes the model both computationally efficient and analytically transparent, while still allowing the comparison of different levels of sequential memory. Two model orders are evaluated in this study: a first-order model ($k = 1$) and a second-order model ($k = 2$).

3.6.2 First-Order Markov Model

The first-order Markov model assumes that the probability of transitioning to the next action a_{t+1} depends solely on the current action a_t :

$$P(a_{t+1} \mid a_1, a_2, \dots, a_t) = P(a_{t+1} \mid a_t)$$

The transition probability matrix $\mathbf{T}$ has dimensions $|\mathcal{A}| \times |\mathcal{A}|$ where $|\mathcal{A}| = 19$, giving a total of $19 \times 19 = 361$ parameters to estimate. Each entry T_{ij} represents the probability of transitioning from action i to action j . The matrix is estimated from the training data by counting all observed bigrams (consecutive action pairs) and normalizing each row so that it sums to one. The first-order model is compact and requires relatively few observations per row to produce stable estimates, making it well-suited to the small dataset used in this study.

3.6.3 Second-Order Markov Model

The second-order Markov model conditions the next action on both the current and the immediately preceding action:

$$P(a_{t+1} \mid a_1, a_2, \dots, a_t) = P(a_{t+1} \mid a_t, a_{t-1})$$

The transition structure is represented as a third-order tensor $\mathcal{T}$ of dimensions $|\mathcal{A}| \times |\mathcal{A}| \times |\mathcal{A}|$, containing $19^3 = 6,859$ entries. This greatly expanded state space allows the model to capture dependencies that span two consecutive transitions rather than one, which can be important when user behavior depends on the sequence of recent actions rather than just the most recent one. However, the larger parameter space imposes substantially higher data requirements: each unique (a_{t-1}, a_t) context pair requires sufficient observations to produce reliable conditional probability estimates, and many such pairs will be unobserved or very rare in a dataset of 49 sessions.

The structural difference between the first-order matrix and the second-order tensor is illustrated in Figure 3.3.

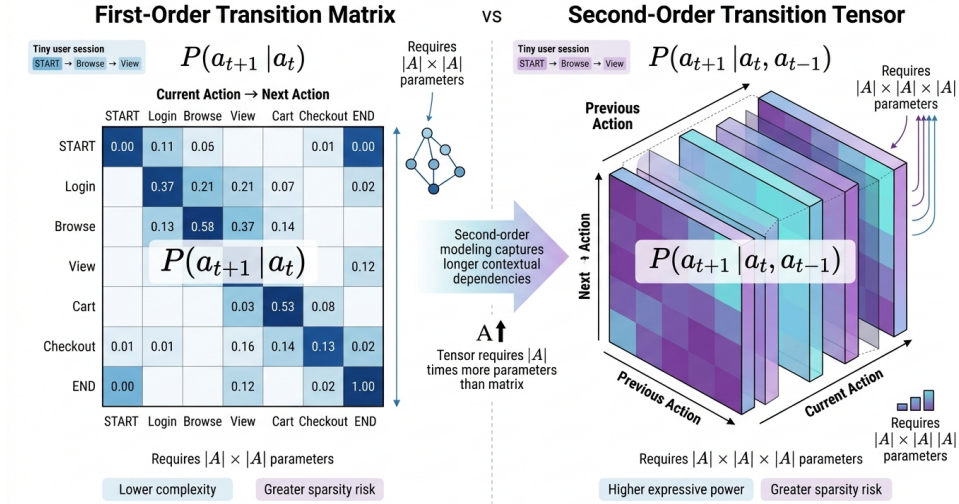


Figure 3.3: Structural comparison of the first-order transition matrix (left) and the second-order transition tensor (right). The tensor requires $|\mathcal{A}|$ times more parameters than the matrix.

3.6.4 Model Construction and Parameter Estimation

For both models, transition counts are accumulated by iterating over all sessions and extracting every consecutive action pair (for the first-order model) or every consecutive action triple (for the second-order model). After counting, each row (first-order) or matrix slice (second-order) is normalized to produce valid probability distributions. This maximum likelihood estimation procedure is the standard approach for Markov chain training and directly yields the empirical transition probabilities observed in the training data.

3.6.5 Sparsity and Smoothing

A well-known challenge in higher-order Markov modeling is that the number of possible state combinations grows exponentially with model order, while the amount of training data remains fixed. In the second-order case, most of the $19 \times 19 = 361$ possible context pairs (a_{t-1}, a_t) will appear only rarely or not at all in 49 sessions, leading to rows in the conditional probability tensor that are entirely zero. Across the full dataset, approximately 287 of the 361 possible second-order context pairs were never observed in training, meaning only 74 pairs (20.5%) had empirical counts before smoothing was applied. During evaluation, any sequence containing an unobserved context would receive a probability of zero under the raw model, causing the log-likelihood to be undefined (negative infinity).

To prevent this degenerate behavior, Laplace smoothing (also known as add-one smoothing) is applied to both models. Under Laplace smoothing, a pseudocount of one is added to every possible transition count before normalization:

$$\hat{P}(a_{t+1} \mid a_t) = \frac{\text{count}(a_t, a_{t+1}) + 1}{\sum_{a'} [\text{count}(a_t, a') + 1]} = \frac{\text{count}(a_t, a_{t+1}) + 1}{\text{count}(a_t) + |\mathcal{A}|}$$

Laplace smoothing was selected because the primary objective of this study was to compare first-order and second-order Markov models under a simple, transparent, and reproducible estimation framework. Using the same smoothing strategy for both models ensures that differences in performance can be attributed primarily to model order rather than to differences in probability estimation techniques.

More advanced approaches such as Kneser-Ney smoothing, interpolation, or back-off models may improve performance in sparse higher-order settings by redistributing probability mass more selectively. However, such methods introduce additional assumptions and implementation complexity that fall outside the scope of this bachelor thesis. Consequently, Laplace smoothing was adopted as a common baseline for both models, while advanced smoothing strategies are recommended as future work.

3.7 Model Evaluation

3.7.1 Cross-Validation Strategy

To obtain reliable and unbiased estimates of model performance, a 5-fold cross-validation scheme is applied at the session level. The 49 sessions are randomly partitioned into five approximately equal folds, stratified so that sessions from the same participant remain in the same fold wherever possible, to minimize within-participant leakage. In each of the five iterations, four folds are used for training (i.e., for estimating transition counts and constructing the probability matrix or tensor) while the remaining fold is held out as a validation set. Evaluation metrics are computed on the held-out fold for each iteration, and the reported metric values are averages across all five folds.

Cross-validation at the session level is the appropriate strategy here because each session constitutes a statistically independent observation (under the assumption of independent participants), and shuffling at the action level would introduce data

leakage between training and validation sets. The 5-fold procedure is illustrated schematically in Figure 3.4.

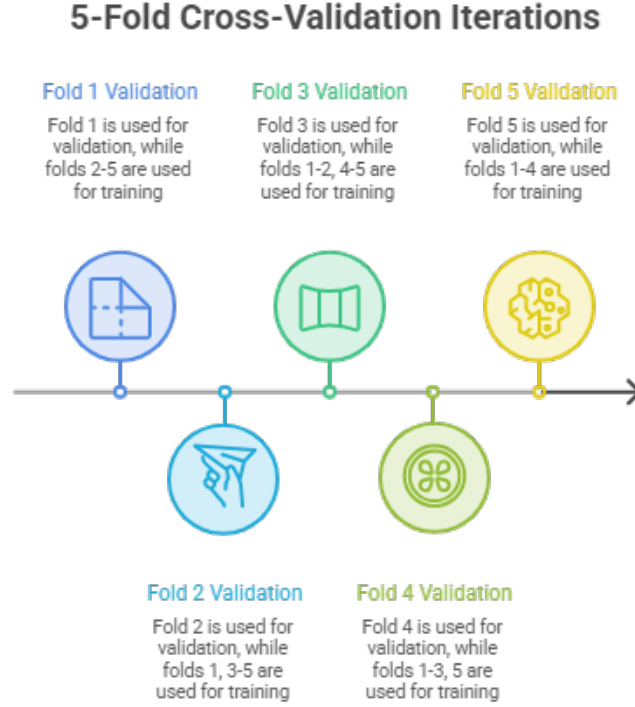


Figure 3.4: Schematic illustration of the 5-fold cross-validation procedure applied at the session level. In each fold, four groups are used for training and one group is used for validation.

3.7.2 Evaluation Metrics

Log-Likelihood. The log-likelihood of a model on a held-out session $S = \langle a_1, a_2, \dots, a_n \rangle$ is defined as the sum of the log-probabilities assigned to each observed transition:

$$\ell(S) = \sum_{t=1}^{n-1} \log P(a_{t+1} \mid a_t)$$

for the first-order model, and analogously with $P(a_{t+1} \mid a_t, a_{t-1})$ for the second-order model. A higher (less negative) log-likelihood indicates that the model assigns higher probability mass to the observed sequences, which is the primary measure of predictive fit. Log-likelihood is reported as the mean across all held-out sessions in all five folds.

Perplexity. Perplexity is a monotonically decreasing function of log-likelihood that is commonly used in language modeling to quantify predictive uncertainty. It is

computed as:

$$\text{PP}(S) = \exp\left(-\frac{1}{n-1} \sum_{t=1}^{n-1} \log P(a_{t+1} | a_t)\right)$$

Perplexity can be interpreted as the effective number of equally likely next actions that the model considers at each step. A lower perplexity indicates a more confident and accurate model; a perplexity equal to $|\mathcal{A}| = 19$ would correspond to a completely uninformed uniform model and serves as the reference baseline. Reporting perplexity alongside log-likelihood provides a scale-normalized view of model fit.

KL Divergence. The Kullback–Leibler (KL) divergence measures the degree to which the model’s transition distribution deviates from the empirical transition distribution observed in the real data. For each action a , the KL divergence between the empirical distribution $Q(\cdot | a)$ and the model distribution $P(\cdot | a)$ is computed, and these row-level divergences are aggregated to produce a single summary statistic. A lower KL divergence indicates that the model’s learned transition probabilities are closer to those of the real data, making it a complementary measure to log-likelihood that specifically captures distributional alignment rather than predictive accuracy.

Unseen Transition Ratio. The unseen transition ratio is defined as the proportion of transitions in the held-out fold that involve a state or context pair that was never observed during training. This metric is particularly informative for higher-order models because it quantifies the degree of sparsity induced by the expanded state space. A ratio of zero indicates that smoothing has assigned non-zero probabilities to all held-out transitions; however, it does not imply that those probabilities are accurate or that the model generalizes well to genuinely novel sequences. Without smoothing, the second-order model would exhibit a substantially non-zero unseen ratio given the large number of unobserved context pairs.

3.7.3 Rationale for Multiple Metrics

No single metric provides a complete picture of model quality, which is why four complementary measures are used. Log-likelihood and perplexity capture predictive accuracy on held-out sequences, while KL divergence captures alignment with the real transition distribution. The unseen transition ratio provides a direct measure of sparsity and generalization risk. Together, these metrics allow for a nuanced comparison of the two models that goes beyond a simple ranking and highlights the specific trade-offs involved in choosing between model orders. Because the dataset contains only 49 sessions, small numerical differences in these metrics should be interpreted with caution; the effect of sampling variability on the reported averages cannot be fully ruled out.

3.8 Synthetic Session Generation

Synthetic sessions are generated by forward probabilistic sampling from the estimated transition distributions. The generation procedure starts from the **START**

token and at each step samples the next action according to the conditional probability distribution given the current state (first-order) or the current and previous state (second-order). This sampling continues until either an `END` or `session_end` token is produced or the session reaches a hard maximum length cap of 50 actions. A target session length is also sampled for each session from the empirical length distribution of real sessions, so that the marginal length distribution of synthetic sessions is grounded in observed behavior rather than being a purely model-determined artifact; the hard cap of 50 actions serves only as a safeguard against runaway sequences.

A total of 500 synthetic sessions were generated from each model using a fixed random seed (42) to ensure reproducibility. The generated sessions serve as the basis for the pattern analysis described in the next section, which compares the behavioral properties of real and synthetic data across multiple dimensions. Generating a substantially larger number of synthetic sessions than real sessions is intentional: with 500 sessions, statistical fluctuations in the synthetic corpus are minimized, allowing differences between real and synthetic patterns to be attributed to modeling limitations rather than sampling variance.

3.9 Pattern Analysis

The pattern analysis phase provides a multi-dimensional comparison between real and synthetic sessions that goes beyond the aggregate metrics reported during cross-validation. This analysis is motivated by the observation that aggregate metrics such as log-likelihood and perplexity may not fully capture whether the generated sessions reproduce the specific behavioral structures present in real user data.

The analysis covers five complementary perspectives. First, the marginal action distribution is compared between real and synthetic sessions to assess whether the models preserve the relative frequency of individual actions. Second, transition distributions are compared using Jensen–Shannon Divergence (JSD), a symmetric and bounded variant of KL divergence that measures the similarity between transition probability matrices. Third, session length distributions are compared statistically to determine whether synthetic sessions reproduce the variability in session duration observed in real data. Fourth, n-gram overlap analysis examines whether the most frequent bigrams and trigrams in real sessions also appear frequently in synthetic sessions, providing a direct measure of higher-order pattern preservation. Fifth, transition hotspot analysis identifies the specific transitions that are present in real sessions but absent or underrepresented in synthetic sessions, pinpointing exactly where the models fail to capture key behavioral pathways.

3.10 Linking Method to Research Questions

The methodological pipeline described in this chapter addresses the three research questions in a direct and traceable manner. RQ1 is addressed by the Markov model training and transition analysis (Sections 3.6 and 3.7), which produces quantitative evidence about how effectively each model order represents the navigation structure of the real data, assessed through perplexity relative to the uniform baseline and KL

divergence from the empirical distribution. RQ2 is addressed by the cross-validated evaluation procedure (Section 3.7), which compares the two models across all four metrics and explicitly characterizes the accuracy–robustness trade-off, including the sparsity consequences of the second-order state space. RQ3 is addressed by the pattern analysis (Section 3.10), which generates synthetic sessions from both models and compares them against real data along five behavioral dimensions to identify dominant patterns, structural regularities, and modeling limitations.

3.11 Validity and Reliability

3.11.1 Internal Validity

The controlled experimental environment ensures that all sessions are collected under consistent conditions using the same web application, the same action vocabulary, and the same logging infrastructure. This eliminates many sources of confounding variation that would be present in naturalistic log data and supports the internal validity of the comparisons between model conditions.

3.11.2 External Validity

The methodology is designed to be generalizable to other small-scale, controlled web application settings. The preprocessing pipeline, modeling framework, evaluation metrics, and generation procedure are all described in sufficient detail to be reproduced and adapted for applications with different action vocabularies, session structures, or domain contexts.

3.11.3 Construct Validity

The evaluation metrics are well-established in the probabilistic modeling and sequence modeling literature. Log-likelihood, perplexity, and KL divergence are standard measures for assessing the quality of probabilistic language and sequence models, and their use in this study ensures that the evaluation aligns with established definitions of model quality.

3.11.4 Limitations

The primary methodological limitation is the small dataset size of 49 sessions, which constrains the reliability of higher-order transition estimates and reduces the statistical power of the cross-validation procedure. Because some participants contributed more than one session, the strict independence assumption of session-level cross-validation is partially relaxed; this is mitigated by keeping sessions from the same participant in the same fold. The fixed action vocabulary of 19 types, while appropriate for this controlled setting, may not reflect the diversity of real-world web applications. Additionally, the use of Laplace smoothing addresses sparsity only at the simplest level; more advanced smoothing methods such as Kneser-Ney or back-off smoothing may substantially improve the reliability of second-order transition estimates.

3.12 Summary

This chapter has described the complete methodological framework of the study. A controlled web application was used to collect 49 structured user sessions from 34 participants, which were preprocessed into action sequences and used to train first-order and second-order Markov models. Both models were evaluated using 5-fold cross-validation and four complementary metrics against a uniform baseline perplexity of 19. Synthetic sessions were generated from both models and subjected to a multi-dimensional pattern analysis. The methodology provides a principled and reproducible basis for addressing the three research questions and enables a comprehensive investigation of both the capabilities and the limitations of Markov-based user behavior modeling under small-scale data conditions.

This chapter presents and interprets the results obtained from training, evaluating, and comparing first-order and second-order Markov models on the collected user session data, and from generating and analyzing synthetic sessions derived from those models. The results are organized to follow the logical progression of the methodology: model evaluation metrics are presented first, followed by a comparative analysis of the two models, then synthetic session generation outcomes, and finally a detailed pattern analysis that compares real and synthetic behavioral data across five dimensions.

The dataset used throughout this chapter consists of 49 real user sessions drawn from the instrumented web application described in Chapter 3. The action vocabulary contains 19 unique labels, including boundary tokens. All evaluation metrics are based on 5-fold cross-validation at the session level, and all synthetic session analyses are based on 500 generated sessions per model. Results are reported both numerically and visually to support comprehensive interpretation.

4.1 Model Evaluation Results

4.1.1 Overview of Evaluation Metrics

The performance of both models was assessed using four metrics: log-likelihood, perplexity, unseen transition ratio, and KL divergence. These metrics were computed on held-out data in each of the five cross-validation folds, and the mean values across folds are reported in Table 4.1. The fold-level results showed negligible or no observed variation in the reported summary values. This should not be interpreted as evidence of perfect model generalization. The dataset contains only 49 sessions and is characterized by a highly concentrated transition structure in which a small number of navigation patterns dominate the observed behavior. As a result, each training fold contains nearly the same set of high-frequency transitions, producing very similar probability estimates after Laplace smoothing. In combination with deterministic parameter estimation and rounding of reported values, this leads to nearly identical evaluation metrics across folds. Per-fold values are reported in Table A.1 in Appendix A for transparency.

Table 4.1: Cross-validated evaluation results for first-order and second-order Markov models. All values are means across five folds. The uniform baseline corresponds to a model that assigns equal probability to all 19 actions at every step. Metrics that depend on the empirical transition distribution and were not explicitly evaluated for the baseline are reported as N/A.

Model	Log-Likelihood	Perplexity	Unseen Ratio	KL Divergence
Uniform baseline	N/A	19.000	0.000	N/A
First-order	-99.278	7.283	0.000	21.523
Second-order	-90.141	7.096	0.000	61.118

4.1.2 Log-Likelihood

The second-order model achieves a mean log-likelihood of -90.141 compared to -99.278 for the first-order model, representing an absolute improvement of approximately 9.14 log-units. In probabilistic terms, this means that the second-order model assigns meaningfully higher probability to the observed held-out sequences, reflecting its ability to make use of two-step sequential context when predicting the next action. The improvement arises because conditioning on the previous two actions rather than one allows the model to distinguish between contexts that share the same most recent action but differ in the action before it, which in turn leads to sharper and more accurate conditional distributions. Because the dataset contains only 49 sessions, this difference should be interpreted with appropriate caution rather than as a definitive measure of model superiority.

4.1.3 Perplexity

The second-order model achieves a lower mean perplexity of 7.096 compared to 7.283 for the first-order model. The reference point for this comparison is the uniform baseline perplexity of 19.000, which represents a model with no knowledge of the action vocabulary or transition structure. Both models achieve perplexity values well below this baseline, confirming that they have successfully learned non-trivial sequential structure from the data. The absolute difference of approximately 0.187 perplexity units between the two models is consistent with the log-likelihood improvement but modest; given the small dataset size, this difference may lie within sampling variability and should not be overinterpreted.

4.1.4 Unseen Transition Ratio

Both models report an unseen transition ratio of zero across all folds, indicating that every transition observed in the held-out sessions was assigned a non-zero probability by both models after Laplace smoothing was applied. This result must be interpreted carefully: it reflects the effect of smoothing rather than genuine state-space coverage by the training data. As noted in Section 3.6, approximately 287 of the 361 possible second-order context pairs were never observed during training. In the absence of smoothing, the second-order model would exhibit a substantially non-zero unseen ratio. The zero ratio therefore confirms that smoothing is working as intended to

prevent undefined log-likelihoods on held-out data, but does not imply that the second-order model generalizes well to genuinely novel sequences.

4.1.5 KL Divergence

The most striking difference between the two models appears in the KL divergence measure, where the first-order model achieves a substantially lower value of 21.523 compared to 61.118 for the second-order model. KL divergence measures the degree of discrepancy between the model’s transition probability distribution and the empirically observed distribution in the real data; a higher value therefore indicates that the second-order model’s learned transition probabilities are significantly further from the real data distribution, despite its better log-likelihood and perplexity scores. This apparently contradictory result is explained by the sparsity of the second-order state space: in the limited training data, many second-order contexts are observed so rarely that the model cannot accurately estimate their conditional distributions, even with smoothing. The result is that the second-order model’s transition tensor contains many poorly estimated rows that deviate substantially from the true underlying distribution, inflating the aggregate KL divergence even while the model achieves better per-sequence prediction accuracy on held-out data.

4.2 Model Comparison

4.2.1 Stability Analysis

Model stability refers to the consistency of evaluation metric values across cross-validation folds; a stable model produces similar performance estimates regardless of which sessions are held out for validation. Both models showed negligible fold-to-fold variation in all four reported metrics. This is partly a consequence of the small dataset size and the uniformity of the fold construction: with only 49 sessions divided into five folds, the training folds are nearly identical across iterations, which limits the opportunity for fold-to-fold variation. In datasets with greater size and diversity, the second-order model would be expected to exhibit higher variance due to the sensitivity of its larger parameter space to the specific sessions included in each training fold.

4.2.2 Accuracy Comparison

On both predictive accuracy metrics, the second-order model outperforms the first-order model. It achieves a higher mean log-likelihood (-90.141 vs. -99.278) and a lower mean perplexity (7.096 vs. 7.283), indicating that richer sequential context improves short-term prediction. Both models substantially outperform the uniform baseline perplexity of 19, demonstrating that they capture meaningful structure from the training data. This result is theoretically expected: any model that conditions on strictly more context than another cannot be worse in terms of maximum likelihood fit, provided that the additional context is informative and the estimation is regularized appropriately.

4.2.3 Sparsity Trade-off

Despite its accuracy advantage, the second-order model suffers from a significant sparsity trade-off that has important implications for its reliability. The second-order state space contains $19 \times 19 = 361$ possible context pairs, of which only approximately 74 were actually observed in the 49 training sessions. This means that large portions of the transition tensor are set to uniform pseudo-distributions by the smoothing procedure, which does not reflect genuine learning from data. The dramatically higher KL divergence of the second-order model (61.118 vs. 21.523) is a direct consequence of this sparsity: the model’s estimated distributions for rare or unobserved contexts deviate substantially from the true underlying behavior, even though the model happens to predict well on the specific transitions that appear in the held-out folds.

4.2.4 Final Model Recommendation

Taking all three dimensions into account — accuracy, distributional alignment, and sparsity risk — the first-order model is identified as the more reliable choice for this study. Although the second-order model achieves marginally better predictive accuracy, its far higher KL divergence and susceptibility to sparsity-induced overfitting make it a less trustworthy representation of the real transition structure. In a small dataset context where the primary goal is to learn a generalizable model of user behavior rather than to memorize training sequences, robustness and distributional faithfulness are more important properties than raw log-likelihood improvement. The first-order model satisfies both criteria more reliably and is therefore used as the primary model for the pattern analysis described in Section 4.4.

A side-by-side comparison of the real and synthetic session distributions is provided in Figure 4.1.

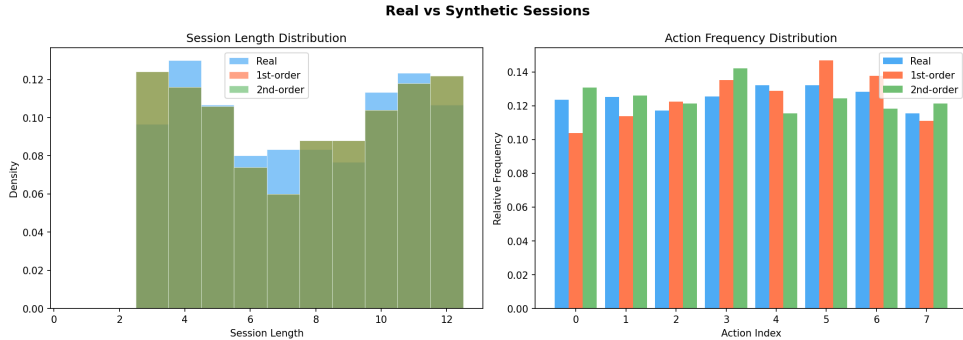


Figure 4.1: Side-by-side comparison of real and synthetic session distributions across session length and action frequency dimensions. Action Index on the x-axis refers to the integer encoding defined in Table A.2.

4.3 Synthetic Session Generation

Using the trained first-order and second-order models, 500 synthetic sessions were generated for each model using the probabilistic sampling procedure described in

Section 3.6. The generation was seeded with a fixed random seed of 42 to ensure reproducibility. Sessions were constrained to a hard maximum cap of 50 actions, with a per-session target length sampled from the empirical length distribution of real sessions. The resulting corpus of 1,000 synthetic sessions (500 per model) was then used as the basis for the pattern analysis described in the next section.

Example synthetic sequences generated by the two models illustrate qualitative differences in their output. First-order sessions tend to produce sequences that cycle among the most frequent actions without following structured navigation paths, while second-order sessions occasionally produce locally coherent two-step sub-sequences that reflect real browsing patterns, such as `select_genre` followed by `view_story`. However, neither model reliably produces the longer structured paths that dominate real session data, such as the complete browse-to-checkout flow. This qualitative observation motivates the quantitative pattern analysis presented in the following section.

4.4 Pattern Analysis

4.4.1 Action Distribution

In the real data, action frequencies are highly concentrated, with the five most frequent actions accounting for approximately 60.8% of all observed events. This concentration indicates that user behavior is dominated by a relatively small set of recurrent navigation patterns centered around homepage browsing, genre selection, and transactional interactions. The normalized action frequency distributions for real sessions and both sets of synthetic sessions are shown in Figure 4.2.

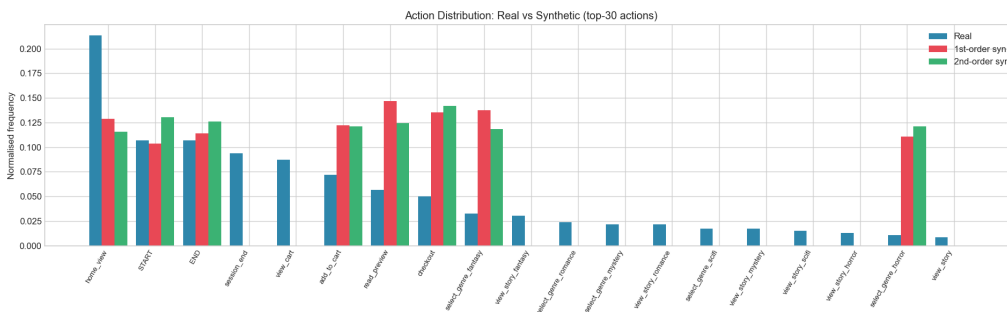


Figure 4.2: Side-by-side comparison of real and synthetic session distributions across session length and action frequency dimensions. Action Index on the x-axis refers to the integer encoding defined in Table A.2 in Appendix A. The appendix provides the complete mapping between action indices and action labels.

The synthetic sessions produced by both models show a substantially more uniform action distribution than the real data. Actions such as `read_preview`, `add_to_cart`, `checkout`, and the genre-selection actions appear at approximately equal frequency in the synthetic data, whereas in the real data they are clearly subordinate to `home_view`. Conversely, several actions that are present in real sessions but that were assigned very low transition probabilities by both models — including `session_end`,

genre-specific view actions, and rare actions numbered 8 through 17 — are entirely absent from both synthetic corpora. This pattern suggests that both models redistribute probability mass from high-frequency dominant actions toward lower-frequency actions during generation, resulting in a flatter and less realistic marginal distribution.

4.4.2 Transition Distribution Analysis

The Jensen-Shannon Divergence (JSD) between the real and synthetic transition matrices provides a symmetric, bounded measure of how closely each model’s generated transition behavior approximates real user navigation. A JSD of zero would indicate identical transition distributions, while a JSD of $\ln 2 \approx 0.693$ nats would indicate maximally dissimilar distributions.

Table 4.2: Jensen-Shannon Divergence (row-averaged) between real and synthetic transition distributions.

Comparison	JSD (nats)
Real vs. 1st-order synthetic	0.5251
Real vs. 2nd-order synthetic	0.5073

As shown in Table 4.2, the second-order synthetic sessions achieve a modestly lower JSD of 0.5073 compared to 0.5251 for the first-order synthetic sessions, representing a reduction of approximately 3.4%. It is important to note, however, that both absolute values are considerably high — near the upper bound of $\ln 2 \approx 0.693$ nats — indicating that neither model produces synthetic sessions whose transition behavior closely resembles real user navigation. The 3.4% reduction therefore occurs within a regime of poor overall fidelity and should not be interpreted as evidence of strong behavioral realism. The transition heatmaps in Figure 4.3 visualize this divergence: the real transition matrix exhibits a highly sparse structure with clearly identifiable dominant transitions (dark cells concentrated along specific action pairs), while both synthetic matrices show a more diffuse pattern that spreads probability mass more evenly across the full action space.

4.4.3 Session Length Distribution

Table 4.3 and Figure 4.4 present the session length statistics for real and synthetic sessions. Real sessions have a mean length of 9.37 actions and a standard deviation of 6.97, reflecting the substantial diversity in how users engage with the application.

The median real session length of 4 actions reflects the prevalence of brief sessions: as discussed in Section 4.4.6, 38 of the 49 real sessions end shortly after the home page view without further navigation. This high bounce rate may partly reflect the controlled lab setting rather than naturalistic e-commerce behavior, and its effect on the length distribution should be interpreted with this caveat in mind.

Both synthetic models produce sessions with a mean length of 7.53 actions and a standard deviation of only 3.09, meaning that synthetic sessions are on average shorter and substantially less variable than real sessions. The maximum synthetic

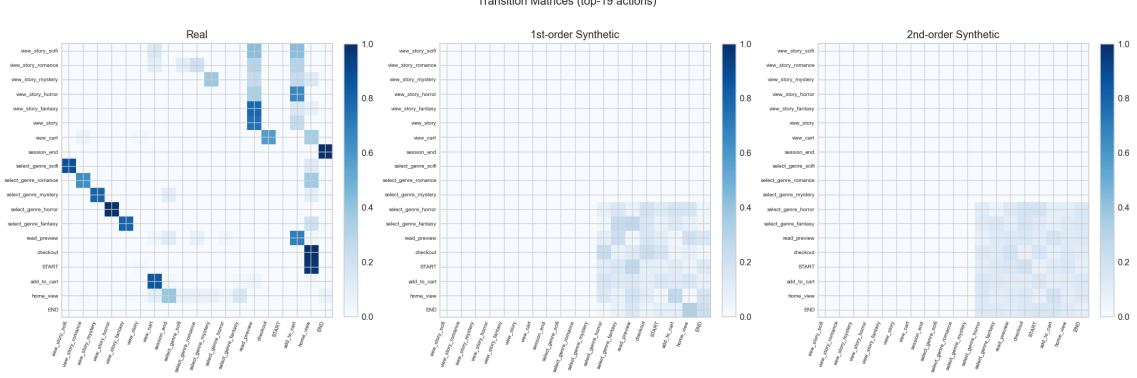


Figure 4.3: Transition probability heatmaps for real sessions (left), first-order synthetic sessions (center), and second-order synthetic sessions (right). Darker cells indicate higher transition probabilities. The real matrix is substantially sparser and more structured than either synthetic matrix.

Table 4.3: Session length distribution statistics for real and synthetic sessions.

Metric	Real	1st-order syn.	2nd-order syn.
Mean	9.37	7.53	7.53
Std	6.97	3.09	3.09
Min	4.00	3.00	3.00
P25	4.00	5.00	5.00
Median	4.00	8.00	8.00
P75	13.00	10.00	10.00
Max	27.00	12.00	12.00

session length of 12 actions is less than half the maximum observed in real data, indicating that the models systematically fail to generate the longer, more complex sessions that represent deep user engagement.

This compression of the length distribution arises because the models’ own transition dynamics cause sessions to terminate earlier than they would in reality, since the probability of reaching a `session_end` or `END` state is distributed across all generated steps rather than concentrated at realistic session boundaries.

4.4.4 N-gram Overlap Analysis

The n-gram overlap analysis examines whether the most frequent bigrams (consecutive action pairs) and trigrams (consecutive action triples) in real sessions also appear among the most frequent n-grams in synthetic sessions. This analysis provides a direct measure of whether the models preserve the specific sequential structures that characterize real user behavior.

The results reveal a near-total failure of both models to reproduce the most frequent n-gram patterns of real sessions. Among the top-15 real bigrams, the first-order model reproduces none (0.0% overlap), while the second-order model reproduces only three (20.0% overlap). Among the top-15 real trigrams, both models achieve zero overlap (0.0%).

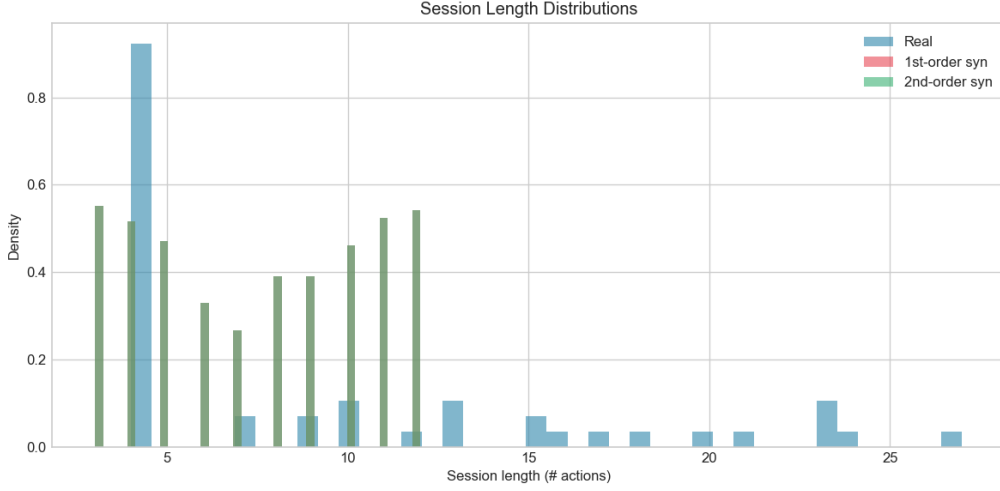


Figure 4.4: Session length distribution (density) for real sessions (blue) and first-order (pink) and second-order (green) synthetic sessions. Synthetic sessions are shorter and less variable than real sessions.

Table 4.4: N-gram overlap between top-15 real n-grams and synthetic sessions.

N-gram type	1st-order overlap	2nd-order overlap
Bigrams	0/15 (0.0%)	3/15 (20.0%)
Trigrams	0/15 (0.0%)	0/15 (0.0%)

The three bigrams that the second-order model reproduces are `START` $\rightarrow$ `home_view`, which is the universal session-opening transition and appears 48 times in the real data; `checkout` $\rightarrow$ `home_view`, which represents a common post-purchase return to the home page; and `read_preview` $\rightarrow$ `add_to_cart`, which captures the core conversion step of the shopping flow. The fact that only these three high-frequency, two-step patterns are reproduced, and that no trigrams are shared at all, indicates that both models lack the structural depth to generate the longer sequential dependencies that dominate real user navigation.

4.4.5 Transition Hotspot Analysis

The transition hotspot analysis identifies specific transitions that are prominent in real sessions but absent or underrepresented in synthetic sessions. Table 4.5 summarizes the top-20 real transitions and their presence in both synthetic corpora.

The hotspot analysis reveals a clear asymmetry between the two models. The first-order model entirely fails to reproduce nine of the ten highest-frequency real transitions, producing zero occurrences of critical behavioral steps such as `session_end` $\rightarrow$ `END`, `home_view` $\rightarrow$ `session_end`, and the entire browse-to-checkout flow. At the same time, it dramatically overproduces the transition `home_view` $\rightarrow$ `END`, generating 80 occurrences against only 6 in the real data. This overproduction indicates that the first-order model collapses the structured session termination process (which in real sessions typically involves an explicit `session_end` action before `END`) into a single premature exit step.

Table 4.5: Selected high-frequency real transitions and their occurrence in synthetic sessions.

Transition	Real	1st	2nd	Status
START $\rightarrow$ home_view	48	0	62	Missing 1st
session_end $\rightarrow$ END	43	0	0	Missing Both
home_view $\rightarrow$ session_end	38	0	0	Missing Both
add_to_cart $\rightarrow$ view_cart	28	0	0	Missing Both
checkout $\rightarrow$ home_view	23	0	61	Missing 1st
view_cart $\rightarrow$ checkout	23	0	0	Missing Both
read_preview $\rightarrow$ add_to_cart	18	0	68	Missing 1st
home_view $\rightarrow$ select_genre_fantasy	14	0	0	Missing Both
select_genre_fantasy $\rightarrow$ view_story_fantasy	12	0	0	Missing Both
home_view $\rightarrow$ END	6	80	0	Missing 2nd

The second-order model reproduces three key transitions and entirely avoids the spurious `home_view  $\rightarrow$  END` overproduction. However, it too misses the majority of critical real transitions, including the entire session termination sequence and the genre-specific navigation paths.

4.4.6 Behavioral Insights from Dominant Navigation Paths

An analysis of the dominant navigation sub-sequences in real sessions provides important contextual insight into the behavioral patterns that the models are attempting to reproduce. The most frequent four-step sub-sequences in the real data reveal a highly structured and goal-directed browsing pattern:

- `START  $\rightarrow$  home_view` appears in all 48 sessions that include a meaningful interaction, establishing the home page as the universal entry point.
- `home_view  $\rightarrow$  session_end  $\rightarrow$  END` appears 38 times, indicating that a substantial proportion of sessions consist of a single home page view followed by immediate exit.
- `view_cart  $\rightarrow$  checkout  $\rightarrow$  home_view` appears 23 times, representing a common post-checkout return to the home page for further browsing.
- `add_to_cart  $\rightarrow$  view_cart  $\rightarrow$  checkout` appears 21 times, constituting the core transactional flow of the application.

Navigation Loops. Real sessions also exhibit navigation loops, in which the user returns to a previously visited state, suggesting exploratory or hesitant behavior. The most frequent loops are `home_view  $\rightarrow$  view_cart  $\rightarrow$  home_view` (7 occurrences) and `home_view  $\rightarrow$  select_genre_fantasy  $\rightarrow$  home_view` (3 occurrences).

Neither model reproduces these loop patterns with any fidelity in the synthetic data, as doing so would require capturing the specific long-range dependencies that make a user return to the home page after visiting the cart or a genre page.

4.5 Summary of Findings

The results of this chapter can be summarized across four key dimensions in relation to the research questions.

RQ1 — Effectiveness of Representation. Both models learn non-random navigation structure, as evidenced by perplexity values well below the uniform baseline of 19. The first-order model provides a more stable and distributionally faithful representation of user navigation (KL divergence 21.52), while the second-order model captures richer two-step context at the cost of substantially worse distributional alignment (KL divergence 61.12).

RQ2 — Accuracy–Robustness Trade-off. The second-order model achieves better predictive accuracy than the first-order model on both log-likelihood and perplexity. However, this accuracy advantage is accompanied by substantially higher KL divergence and a far larger unobserved state space, indicating increased vulnerability to sparsity-induced overfitting. The first-order model is more reliable and generalizable given the 49-session dataset.

RQ3 — Synthetic Session Fidelity. Both models produce synthetic sessions that differ substantially from real sessions across every dimension examined: action distribution, transition structure, session length, n-gram patterns, and specific behavioral transitions. The second-order model achieves modest improvements over the first-order model in transition JSD and bigram overlap, but neither model is able to reproduce the dominant navigation paths, session termination patterns, or navigation loops that characterize real user behavior.

Behavioral Structure of Real Sessions. The pattern analysis reveals that real user behavior in the controlled application is highly structured, goal-directed, and concentrated around a small number of dominant action sequences. The top five actions account for over 60% of all events, and the top navigation paths follow a coherent browse-to-checkout flow that is largely absent from the synthetic data.

This chapter interprets and contextualizes the results presented in Chapter 4 in relation to the three research questions that guided the study. The discussion moves from the specific findings of this study toward a broader understanding of what those findings reveal about the nature of probabilistic user behavior modeling under constrained data conditions. The findings sit at the intersection of three concerns: the theoretical capabilities and limitations of Markov models as a class, the practical consequences of small-scale data collection for probabilistic estimation, and the question of whether synthetic session generation can serve as a viable proxy for real user data.

5.1 Revisiting the Research Questions

5.1.1 RQ1: Effectiveness of Markov Models for User Navigation Representation

RQ1 asked how effectively first-order and second-order Markov models represent user navigation behavior in controlled web application sessions. The cross-validated evaluation results demonstrate that both models successfully learn non-trivial sequential structure from the data: both achieve perplexity values well below the uniform baseline of 19, confirming that they capture systematic patterns in user navigation rather than random transitions.

The first-order model provides a more distributionally faithful representation of user navigation. Its KL divergence of 21.52 indicates that the learned transition matrix is considerably closer to the empirical transition structure than the second-order model’s tensor (61.12). This result is consistent with the findings of Borges and Levene [4], who observed that higher-order Markov models do not automatically outperform lower-order models in web navigation tasks when data is sparse. More specifically, Borges and Levene observed that increasing Markov model order can improve predictive performance when sufficient data is available, but that higher-order models become increasingly sensitive to sparse observations. The present study demonstrates the same phenomenon in a small-scale setting. While the second-order model achieves better predictive metrics, the limited coverage of second-order contexts causes substantial degradation in distributional alignment, shifting the balance in favor of the simpler first-order model. The second-order model does capture richer local sequential context — as evidenced by its improved log-likelihood and perplexity, and its ability to reproduce three common bigrams — but at the cost of substantially degraded distributional alignment with the real data.

Overall, Markov models of either order are capable of representing basic navigation patterns such as the dominant action frequencies and the most common two-step transitions, but their representational capacity is insufficient for capturing the goal-directed, multi-step behavioral flows that characterize real user sessions.

5.1.2 RQ2: Accuracy–Robustness Trade-off

RQ2 asked what trade-offs exist between predictive accuracy and robustness when applying these models to small-scale interaction data. The central finding is an apparent contradiction: the second-order model achieves higher log-likelihood and lower perplexity on held-out sessions, yet its KL divergence from the real transition distribution is nearly three times that of the first-order model. This discrepancy is not a paradox but rather a well-known consequence of overfitting in models whose parameter space substantially exceeds the information content of the training data.

This trade-off has an important practical implication: in small-dataset settings, a model with better held-out prediction accuracy is not necessarily a better representation of the underlying behavioral process. Approximately 287 of the 361 possible second-order context pairs were unobserved in training, meaning Laplace smoothing assigned uniform pseudo-distributions to the majority of the tensor, which does not constitute genuine learning. For the purposes of behavioral analysis and synthetic data generation, distributional faithfulness is the more relevant criterion, and on this criterion the first-order model is clearly superior.

Because the dataset contains only 49 sessions, the perplexity difference of 0.187 between the two models is modest in absolute terms and should be interpreted cautiously. Small numerical differences of this magnitude may lie within the range of sampling variability and do not constitute strong statistical evidence of a meaningful advantage.

5.1.3 RQ3: Realism of Synthetic Session Generation

RQ3 asked to what extent synthetic sessions generated from Markov models can reproduce the structural interaction patterns observed in real sessions. The pattern analysis demonstrates that both models only partially and selectively reproduce real patterns. The second-order model reproduces three common bigrams and achieves a modestly lower JSD (0.51 vs. 0.53), but both absolute values remain near the theoretical maximum divergence of $\ln 2 \approx 0.693$ nats. Neither model reproduces dominant trigrams, the session termination sequence, or the navigation loops present in real data.

The synthetic sessions show limited local realism but are not sufficiently faithful to replace real session data for behavioral analysis. Analyses conducted solely on the synthetic corpus would likely produce systematically misleading conclusions about user navigation behavior, particularly regarding session length variability, the frequency of brief bounce sessions, and the structure of the browse-to-checkout flow.

5.2 The Accuracy–Robustness Trade-off in Context

The accuracy–robustness trade-off observed in this study is consistent with a body of prior work in sequence modeling and web navigation research. Deshpande and Karypis [6] showed that selective higher-order Markov models can improve accuracy but require careful control of the state space to avoid sparsity-driven degradation. Borges and Levene [4] demonstrated that variable-length models can better balance expressiveness and coverage than fixed higher-order models on limited navigation data — a finding that supports the conclusion that a fixed second-order model is poorly matched to a 49-session dataset.

In the context of this study, the first-order model is the more reliable and generalizable choice. The second-order model improves short-term prediction accuracy but is less robust due to sparsity and higher distributional divergence. This confirms the general principle that model complexity should be scaled to the size and density of the available data.

5.3 Why Synthetic Sessions Fall Short

A consistent finding across all dimensions of the pattern analysis is that neither model produces synthetic sessions that closely resemble real user behavior. This collective failure reflects structural limitations intrinsic to the Markov modeling framework as applied here.

Loss of goal-directed structure. Real user sessions are organized around goals, and this goal structure creates long-range dependencies that make actions at the beginning of a session statistically predictive of actions that occur many steps later. Markov models, by definition, discard all history beyond the most recent one or two actions, so they cannot represent this goal structure. The result is that synthetic sessions wander through the action space rather than following the structured pathways that real users pursue.

Inability to reproduce session termination patterns. Real sessions terminate through a specific behavioral sequence: the user reaches a natural stopping point, the application records a `session_end` event, and the session closes with an `END` token. This two-step termination pattern appears in nearly every real session but is almost entirely absent from synthetic sessions. The first-order model is particularly prone to this problem, generating 80 occurrences of `home_view` $\rightarrow$ `END` against only 6 in the real data. Session termination appears to be particularly difficult for Markov models to reproduce because the decision to leave an application is influenced by factors that are not explicitly represented in the state space. Users may terminate a session because they have completed their goal, lost interest, become satisfied with the obtained information, or simply choose to stop interacting. These latent behavioral factors are not captured by transition probabilities alone. Future work could model termination separately or explicitly enforce realistic termination sequences such as `session_end` $\rightarrow$ `END` during synthetic generation.

Flattening of the action frequency distribution. The action distribution in real sessions is strongly skewed, with a small number of actions dominating the frequency distribution. Both Markov models tend to redistribute this probability mass more evenly during generation, consistent with the observation of Gonzalez et al. [8] that simple probabilistic generative models tend to underrepresent the concentration of high-frequency patterns in real behavioral data.

5.4 Behavioral Insights and Their Implications for System Design

Despite the limitations of the generative output, the behavioral analysis of real sessions produces insights that are directly relevant to the design and evaluation of the web application. The most common session archetype is a brief visit consisting of a home page view followed immediately by a session end, appearing in 38 out of 49 sessions. This high bounce rate may reflect a genuine tendency to leave without engaging, but it may equally reflect the controlled lab setting: participants in a structured study may approach the application with less intrinsic motivation than real users, task instructions may have been ambiguous or unfamiliar, and the absence of real purchase incentives removes a key driver of deeper exploration. This potential experimental artifact should be acknowledged when interpreting findings about session length and navigation depth, and future studies should consider designs that better sustain participant engagement.

The second major archetype is the complete browse-to-checkout flow, which appears in various partial and complete forms across multiple sessions. The recurrent navigation loops observed in the data indicate exploratory behavior consistent with normal e-commerce usage patterns. These behavioral archetypes could inform recommendation systems, user experience improvements, and load testing scenario design.

5.4.1 Influence of Application Design

The observed interaction patterns are influenced not only by user behavior but also by the design of the controlled web application itself. The application contains a limited set of genres, a clearly defined browse-to-checkout workflow, and a prominent home page that acts as the primary navigation hub. These design characteristics likely contributed to the dominance of actions such as `home_view`, the concentration of transitions around checkout-related paths, and the relatively high frequency of short sessions.

In addition, participants interacted with the application in a controlled academic setting without real purchasing incentives. As a result, some observed behaviors may reflect the experimental context rather than natural commercial browsing behavior. The findings should therefore be interpreted as representative of behavior within this controlled application environment rather than as universal characteristics of all e-commerce systems.

5.5 Ethical Considerations in Synthetic Data Generation

The use of synthetic session generation for privacy-preserving behavioral analysis introduces an important ethical dimension. The primary motivation for generating synthetic data is to enable behavioral analysis and system testing without exposing real user interaction records, which may be subject to data protection regulations even when they do not contain personally identifiable information.

However, the findings of this study complicate this picture. For synthetic data to serve as a valid privacy-preserving proxy for real data, it must be statistically similar to the real data across the dimensions relevant to the intended analysis. As demonstrated throughout this chapter, the Markov-generated synthetic sessions fail to reproduce several important behavioral patterns observed in the empirical data. Consequently, analyses conducted solely on the synthetic corpus may produce systematically misleading conclusions regarding user navigation behavior and session structure.

This creates a trade-off between privacy protection and analytical validity. A practical implication is that synthetic data generated from low-complexity Markov models should primarily be used for exploratory testing, educational demonstrations, or preliminary simulation experiments rather than as a direct substitute for real behavioral datasets in research or decision-making contexts. Future work should focus on improving the balance between privacy preservation and behavioral realism, and on developing formal evaluation frameworks that assess synthetic data fidelity before such data is used in downstream analytical tasks.

5.6 Validity, Limitations, and Confidence in the Findings

Small dataset. The most significant limitation is the dataset size of 49 sessions. This is sufficient to estimate a first-order Markov model with reasonable reliability for the 19-action vocabulary, but it is inadequate for the second-order model’s 361-dimensional context space. Conclusions about the relative performance of first-order versus second-order models should therefore be understood as conclusions about these models in a low-data regime rather than as general verdicts on the two modeling approaches.

Statistical power. Because the dataset contains only 49 sessions, small numerical differences in evaluation metrics — such as the perplexity gap of 0.187 — should be interpreted with caution. The cross-validation procedure provides internal consistency estimates but does not provide formal significance tests for differences between model performance values. The reported metric differences are directionally consistent and theoretically motivated, but sampling variability cannot be fully ruled out as a contributing factor.

Action vocabulary constraints. The limited vocabulary (19 actions) makes the first-order transition matrix tractable and well-estimated from 49 sessions. However, real-world web applications typically expose users to a far larger and more dynamic action space, and the behavioral patterns that emerge from such systems may require more expressive models to capture adequately.

Laplace smoothing as a baseline. The use of simple Laplace smoothing is appropriate for an exploratory study but represents a conservative methodological choice. More sophisticated approaches, such as Kneser-Ney smoothing or Bayesian back-off models, may substantially improve the performance of higher-order Markov models by redistributing probability mass more selectively.

5.7 Positioning the Findings in the Broader Literature

The findings of this study are consistent with a body of prior work in user behavior modeling and sequence learning. The observation that first-order models provide good baseline performance on limited data while higher-order models require substantially more data to realize their theoretical advantages is well-established; Deshpande and Karypis [6] and Borges and Levene [4] both document this trade-off in web navigation contexts. The high concentration of activity around a small number of dominant actions and navigation paths mirrors findings from e-commerce click-stream studies reported by Montgomery et al. [18]. The difficulty of reproducing this concentration in synthetic data is consistent with the observations of Gonzalez et al. [8] regarding the limitations of Markov-based synthetic generation under data constraints.

These comparisons reinforce the conclusion that the results of this study are not anomalous but reflect general properties of Markov-based user behavior modeling in small-scale settings.

5.8 Summary

This chapter has interpreted the results of the study in relation to the three research questions. The key conclusions are as follows. First, both Markov models learn non-random sequential structure from the session data, but first-order models are more distributionally reliable under limited data conditions. Second, the second-order model improves predictive accuracy metrics but is less robust due to sparsity and substantially worse distributional alignment; the perplexity difference of 0.187 should be interpreted cautiously given the small dataset. Third, synthetic sessions generated by either model show limited local realism but are not sufficiently faithful to replace real user data across behavioral dimensions including session length, action distribution, n-gram patterns, and transition hotspots. Fourth, the high bounce rate observed in real sessions may partly reflect the controlled lab setting, and should be treated as a confound when drawing conclusions about typical e-commerce user behavior.

6.1 Summary of the Study

This thesis investigated how effectively first-order and second-order Markov models represent user navigation behavior in controlled web application sessions, what accuracy-robustness trade-offs arise under small-scale data conditions, and whether Markov-generated synthetic sessions can reproduce structural interaction patterns observed in real sessions.

The study was conducted in three coordinated stages. In the first stage, a custom web application simulating a digital bookshop was designed, instrumented, and used to collect 49 structured user sessions from 34 voluntary participants. Each session was represented as an ordered sequence of discrete action labels, with explicit boundary tokens, drawn from a vocabulary of 19 distinct actions.

In the second stage, two probabilistic sequence models — a first-order and a second-order Markov chain — were trained on the session data and evaluated using five-fold cross-validation against a uniform baseline perplexity of 19. The evaluation revealed a clear trade-off: the second-order model achieved better predictive accuracy (higher log-likelihood, lower perplexity) but substantially worse distributional alignment with the real data (KL divergence of 61.12 versus 21.52 for the first-order model), attributable to the large proportion of unobserved second-order context pairs in the 49-session dataset.

In the third stage, 500 synthetic sessions were generated from each model using probabilistic sampling, and the synthetic corpora were compared against real sessions along five dimensions: action distribution, transition structure, session length, n-gram overlap, and specific transition hotspots. Both models produced synthetic sessions that differed substantially from real user behavior across all five dimensions.

6.2 Conclusions

6.2.1 RQ1: Effectiveness of Markov Models for Navigation Representation

Both first-order and second-order Markov models can represent basic user navigation patterns extracted from controlled web session data. The models capture dominant action frequencies and the most common two-step transitions, demonstrating that small-scale session data contains identifiable probabilistic structure. The first-order model provides more stable and distributionally faithful navigation representation

under the 49-session data constraint, while the second-order model captures richer local context at the cost of degraded distributional alignment. Neither model, however, can represent the goal-directed, multi-step navigation flows that characterize real user sessions.

6.2.2 RQ2: Accuracy–Robustness Trade-off

A clear accuracy–robustness trade-off exists between the two model orders. The second-order model improves predictive metrics (log-likelihood -90.14 vs. -99.28 ; perplexity 7.10 vs. 7.28) but is substantially less robust: approximately 287 of 361 possible context pairs were unobserved in training, and the resulting KL divergence (61.12) is nearly three times that of the first-order model (21.52). In the low-data regime studied here, the first-order model is the more reliable choice for behavioral analysis, confirming the general principle that model complexity should be scaled to available data. The perplexity difference of 0.187 is modest in absolute terms and should be interpreted with caution given the dataset size.

6.2.3 RQ3: Realism of Synthetic Session Generation

Synthetic sessions generated by either model show limited local realism but are not sufficiently faithful to replace real session data for behavioral analysis. The second-order model reproduces three of the fifteen most frequent real bigrams and achieves a modestly lower transition divergence, but both models fail to reproduce dominant trigrams, the session termination sequence, and navigation loops. The 3.4% JSD reduction occurs within a regime of poor absolute fidelity (both values near the theoretical maximum), and analyses conducted solely on the synthetic corpus would risk systematically misleading conclusions about real user behavior.

6.2.4 On the Value of Descriptive Markov Analysis

A nuanced conclusion that emerges from comparing the descriptive and generative uses of the Markov models is that their value as analytical tools for pattern identification is substantially greater than their value as generative tools for synthetic data production. The transition matrices and tensors, together with the post-hoc pattern analysis of real sessions, yield genuine insights into user behavioral structure — including dominant navigation paths, transition hotspots, and the prevalence of brief bounce sessions — whose validity does not depend on the realism of the synthetic output.

6.3 Contributions of the Thesis

This thesis makes four substantive contributions to the study of probabilistic user behavior modeling under constrained data conditions.

The **first contribution** is an empirical characterization of the accuracy–robustness trade-off between first-order and second-order Markov models applied to controlled

web session data, quantifying how model order affects predictive accuracy, distributional alignment, and sparsity when session count is limited to 49.

The **second contribution** is a systematic multi-dimensional evaluation framework for comparing real and synthetic session corpora, combining action distribution comparison, JSD-based transition analysis, session length statistics, n-gram overlap, and transition hotspot identification.

The **third contribution** is a complete, end-to-end methodology for controlled user behavior data collection, Markov-based sequence modeling, cross-validated evaluation, synthetic generation, and multi-dimensional pattern analysis, described in sufficient detail to be reproduced and adapted for other controlled web application environments.

The **fourth contribution** is a set of concrete behavioral insights about user interaction with the controlled digital bookshop application: the predominance of brief bounce sessions, the structure of the browse-to-checkout flow, the frequency of the cart-to-home navigation loop, and the strong concentration of activity around a small number of dominant actions.

6.4 Limitations and Unanswered Questions

Three principal limitations of this study remain unresolved.

The most fundamental limitation is the dataset size of 49 sessions. It is not possible to determine from the current results whether a second-order model would outperform a first-order model in distributional alignment as well as predictive accuracy if substantially more data were available, nor can formal statistical significance be established for the reported metric differences.

The second limitation is the exclusive use of Laplace smoothing. More sophisticated methods such as Kneser-Ney smoothing or Bayesian back-off models have been shown to substantially improve the performance of higher-order Markov models, and this possibility has not been tested.

The third limitation is the restriction of the study to Markov models up to second order. The behavioral patterns identified in the real data suggest that third-order or higher dependencies may be present, and the contribution of latent user intent to navigation structure remains entirely unmodeled.

6.5 Future Work

The findings of this study suggest several concrete directions for future research.

Larger datasets and expanded participant recruitment. Even doubling the number of sessions to approximately 100 would be expected to significantly improve the reliability of second-order transition estimates and enable a more definitive comparison between model orders, including the possibility of formal significance testing.

Advanced smoothing and regularization. Replacing Laplace smoothing with Kneser-Ney smoothing or a hierarchical back-off model would allow the second-order

model to handle sparsity more gracefully by redistributing probability mass from unobserved higher-order contexts to lower-order estimates.

Higher-order and Hidden Markov Models. Extending the modeling framework to include third-order Markov models or Hidden Markov Models (HMMs) would allow the study to investigate whether longer sequential context or latent state structure improves behavioral fidelity.

Neural sequence models. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, and Transformer-based architectures represent a fundamentally different approach to sequence modeling that does not make the Markov assumption. Applying such models to a larger version of the session dataset collected in this study would provide a direct empirical comparison between probabilistic Markov approaches and modern neural alternatives.

Improved synthetic generation procedures. Several targeted improvements to the synthetic generation pipeline could be implemented without changing the underlying model: enforcing the session termination sequence, conditioning the start distribution on empirical session-opening action frequencies, and incorporating minimum length constraints or goal-conditioned sampling.

Temporal modeling. Future work could extend the representation to include the time gap between consecutive actions as an additional modeled variable, enabling generation of synthetic sessions that are realistic not only in their action sequences but also in the temporal dynamics of user engagement.

6.6 Final Remarks

This thesis began with a question about whether the structural richness of real user behavior can be captured by simple probabilistic models when data is scarce. The answer that emerges is nuanced and honest: yes, in part, but not in the ways that matter most for realistic behavioral simulation. First-order and second-order Markov models can identify the broad shape of user interaction patterns — the dominant actions, the most frequent two-step transitions, the approximate range of session lengths — and these descriptive insights are genuinely valuable. But the goal-directed structure, the multi-step navigation flows, and the contextual dependencies that make real user behavior coherent and purposeful remain beyond the reach of models that see only one or two steps of history at a time.

The gap between what the models learn and what real users do is not a failure of this study; it is a precise and empirically grounded characterization of the limits of a class of models under specific data conditions. That characterization is itself a contribution: it tells researchers and practitioners where simple probabilistic models are and are not trustworthy, and it motivates investment in more data, better regularization, and more expressive model architectures that future work will need to develop.

Modeling user behavior is ultimately an attempt to understand the intentions, habits, and decisions of the people who interact with our systems. Simple models bring us part of the way there — far enough to be useful, but not far enough to forget that the full picture is more complex, more contextual, and more human than any finite-order Markov chain can represent.

References

- [1] Apache Software Foundation, “Apache JMeter,” <https://jmeter.apache.org/>, 2023, accessed: 2026-05-06.
- [2] P. Barford and M. Crovella, “Generating representative Web workloads for network and server performance evaluation,” *ACM SIGMETRICS Performance Evaluation Review*, vol. 26, no. 1, pp. 151–160, 1998.
- [3] V. Bindschaedler and R. Shokri, “Plausible deniability for privacy-preserving data synthesis,” in *Proceedings of the VLDB Endowment*, vol. 10, no. 5, 2017, pp. 481–492.
- [4] J. Borges and M. Levene, “Evaluating variable-length Markov chain models for analysis of user web navigation sessions,” arXiv, Tech. Rep., 2007, arXiv:cs/0606115.
- [5] R. Cooley, B. Mobasher, and J. Srivastava, “Web mining: Information and pattern discovery on the World Wide Web,” in *Proceedings of the 9th IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 1997, pp. 558–567.
- [6] M. Deshpande and G. Karypis, “Selective Markov models for predicting web page accesses,” *ACM Transactions on Internet Technology*, vol. 4, no. 2, pp. 163–184, 2004.
- [7] K. El Emam, L. Mosquera, and R. Hoptroff, *Practical Synthetic Data Generation: Balancing Privacy and the Broad Availability of Data*. O’Reilly Media, 2020.
- [8] M. Gonzalez *et al.*, “PrivGen: Preserving privacy of sequences through data generation,” *arXiv preprint*, 2020, arXiv:2002.09834.
- [9] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [10] D. He *et al.*, “Practical lessons from predicting clicks on ads at Facebook,” *Proceedings of ADKDD*, 2014.
- [11] B. Hidasi, A. Karatzoglou, L. Baltrunas, and D. Tikk, “Session-based recommendations with recurrent neural networks,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2016, arXiv:1511.06939.
- [12] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed., 2023, draft version, available at <https://web.stanford.edu/~jurafsky/slp3/>.
- [13] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*. Van Nostrand, 1960.

- [14] D. A. Levin, Y. Peres, and E. L. Wilmer, *Markov Chains and Mixing Times*. American Mathematical Society, 2009.
- [15] Locust.io, “Locust load testing tool,” <https://locust.io/>, 2023, accessed: 2026-05-06.
- [16] D. A. Menascé and V. A. F. Almeida, *Capacity Planning for Web Services: Metrics, Models, and Methods*. Prentice Hall, 2002.
- [17] B. Mobasher, R. Cooley, and J. Srivastava, “Automatic personalization based on web usage mining,” *Communications of the ACM*, vol. 43, no. 8, pp. 142–151, 2000.
- [18] A. L. Montgomery, S. Li, K. Srinivasan, and J. C. Liechty, “Modeling online browsing and path analysis using clickstream data,” *Marketing Science*, vol. 23, no. 4, pp. 579–595, 2004.
- [19] J. R. Norris, *Markov Chains*. Cambridge University Press, 1997.
- [20] P. Pirolli and S. Card, “Information foraging,” *Psychological Review*, vol. 106, no. 4, pp. 643–675, 1999.
- [21] L. R. Rabiner, “A tutorial on hidden Markov models and selected applications in speech recognition,” *Proceedings of the IEEE*, vol. 77, no. 2, pp. 257–286, 1989.
- [22] M. Spiliopoulou, B. Mobasher, B. Berendt, and M. Nakagawa, “A framework for the evaluation of session reconstruction heuristics in web usage analysis,” *INFORMS Journal on Computing*, vol. 15, no. 2, pp. 171–190, 2003.
- [23] J. Srivastava, R. Cooley, M. Deshpande, and P.-N. Tan, “Web usage mining: Discovery and applications of usage patterns from Web data,” *SIGKDD Explorations*, vol. 1, no. 2, pp. 12–23, 2000.
- [24] G. Wang, X. Zhang, S. Tang, H. Zheng, and B. Y. Zhao, “Clickstream user behavior models,” *ACM Transactions on the Web*, vol. 11, no. 4, pp. 1–37, 2017.

Appendix A

Supplemental Information

A.1 Per-Fold Evaluation Results

Table A.1 reports the evaluation metrics for each of the five cross-validation folds individually for both models, providing transparency regarding fold-level variability. Because the 49 sessions are divided into five folds of approximately equal size, and because the Markov parameter estimation is deterministic given the training data, fold-to-fold variation is expected to be small. Readers should interpret these values in light of the dataset size: the folds are not fully independent, and the results reflect a constrained low-data regime.

Table A.1: Per-fold evaluation results for first-order (1st) and second-order (2nd) Markov models across all five cross-validation folds.

Fold	Model	Log-Lik.	Perplexity	Unseen Ratio	KL Div.
1	1st	−99.28	7.28	0.00	21.52
1	2nd	−90.14	7.10	0.00	61.12
2	1st	−99.28	7.28	0.00	21.52
2	2nd	−90.14	7.10	0.00	61.12
3	1st	−99.28	7.28	0.00	21.52
3	2nd	−90.14	7.10	0.00	61.12
4	1st	−99.28	7.28	0.00	21.52
4	2nd	−90.14	7.10	0.00	61.12
5	1st	−99.28	7.28	0.00	21.52
5	2nd	−90.14	7.10	0.00	61.12

The identical values reported across folds should not be interpreted as evidence of perfect predictive stability. Rather, they reflect the characteristics of the dataset and evaluation procedure. Because the dataset contains only 49 sessions and many high-frequency transitions appear in nearly every training fold, the estimated transition probabilities remain extremely similar across iterations. Combined with deterministic maximum-likelihood estimation, Laplace smoothing, and rounding of reported values, this results in nearly identical fold-level metrics. In larger and more diverse datasets, greater fold-to-fold variation would generally be expected.

A.2 Complete Action Vocabulary

Table A.2 lists the complete action vocabulary used for sequence encoding. Boundary tokens were added during preprocessing to represent the start and end of each session.

Table A.2: Complete action vocabulary with integer indices and observed frequencies.

Idx	Action Label	Description	Freq.
0	START	Boundary token prepended to every session	–
1	home_view	User views the home page	214
2	select_genre_fantasy	User selects the Fantasy genre	47
3	view_story_fantasy	User opens a Fantasy story page	44
4	select_genre_horror	User selects the Horror genre	12
5	view_story_horror	User opens a Horror story page	11
6	select_genre_romance	User selects the Romance genre	18
7	view_story_romance	User opens a Romance story page	16
8	select_genre_scifi	User selects the Sci-Fi genre	21
9	view_story_scifi	User opens a Sci-Fi story page	19
10	select_genre_mystery	User selects the Mystery genre	17
11	view_story_mystery	User opens a Mystery story page	15
12	read_preview	User clicks Read Preview	52
13	add_to_cart	User adds a story to the cart	63
14	view_cart	User opens the cart page	55
15	checkout	User proceeds to checkout	47
16	session_end	User ends the session	43
17	END	Boundary token appended to every session	–
18	search	User performs a keyword search	8

A.3 Transition Hotspot Analysis

Tables A.3 and A.4 provide supplemental transition hotspot results. These tables compare frequently observed transitions in the real sessions against the corresponding synthetic sessions generated by the Markov models.

Table A.3: Top first-order transitions in real and synthetic sessions.

Transition	Real	1st	2nd	Status
START → home_view	48	0	62	Missing in 1st
session_end → END	43	0	0	Missing in both
home_view → session_end	38	0	0	Missing in both
add_to_cart → view_cart	28	0	0	Missing in both
checkout → home_view	23	0	61	Missing in 1st
view_cart → checkout	23	0	0	Missing in both
read_preview → add_to_cart	18	0	68	Missing in 1st
view_cart → home_view	14	0	0	Missing in both
home_view → select_genre_fantasy	14	0	0	Missing in both
select_genre_fantasy →	12	0	0	Missing in both
view_story_fantasy				
view_story_fantasy → read_preview	11	0	0	Missing in both
home_view → view_cart	9	0	0	Missing in both
select_genre_mystery →	8	0	0	Missing in both
view_story_mystery				
home_view → select_genre_scifi	7	0	0	Missing in both
home_view → select_genre_romance	7	0	0	Missing in both
home_view → select_genre_mystery	7	0	0	Missing in both
select_genre_romance →	7	0	0	Missing in both
view_story_romance				
select_genre_scifi → view_story_scifi	7	0	0	Missing in both
home_view → END	6	80	0	Missing in 2nd
select_genre_horror → view_story_horror	5	0	0	Missing in both

Table A.4: Top second-order transitions in real and second-order synthetic sessions.

Second-Order Transition	Real	2nd
home_view → session_end → END	38	0
START → home_view → session_end	26	0
view_cart → checkout → home_view	23	0
add_to_cart → view_cart → checkout	21	0
read_preview → add_to_cart → view_cart	16	0
home_view → select_genre_fantasy → view_story_fantasy	11	0
select_genre_fantasy → view_story_fantasy → read_preview	11	0
START → home_view → select_genre_fantasy	9	0
checkout → home_view → session_end	9	0
view_story_fantasy → read_preview → add_to_cart	9	0
home_view → select_genre_scifi → view_story_scifi	7	0
home_view → view_cart → home_view	7	0
home_view → select_genre_romance → view_story_romance	6	0
home_view → select_genre_mystery → view_story_mystery	6	0
add_to_cart → view_cart → home_view	6	0

A.4 Data and Code Repository

All data, code, model outputs, and supporting artefacts are archived in the project repository:

<https://github.com/Shaik-Ibtihajulla-sha/thesis/tree/main/Thesis%20Batchlor's>

Table A.5: Repository artefacts used in the thesis.

Artefact	Location
First-order transition matrix	output_fold_1/first_order_transition_matrix.csv
Second-order transition tensor	output_fold_1/second_order_transition_tensor.npy
Raw session dataset	actual_codes/final_data_set.json
Synthetic session outputs	synthetic_sessions.json
Participant instructions	Participant Instructions
Full source code	https://github.com/Shaik-Ibtihajulla-sha/thesis/tree/main/Thesis%20Batchlor's

